Supplementary Material: VarDictRust: A Memory-Bounded, Oracle-Equivalent Rust Consolidation of the VarDict Variant-Calling Pipeline for Deep Whole-Genome and Ultra-Low-VAF Sequencing

Authors: Palak Sheth[1,*]
[1]Independent Researcher (palakpsheth@gmail.com | https://www.linkedin.com/in/palak-p-sheth/)
Engine Release: `VarDictRust` `v1.0.6` (commit `2a8e9a3`, engine fingerprint `5865a2a0b24a8c17`, performance fingerprint `c490c36317b8f25b`)
Reference Oracles: `VarDictJava v1.8.3 (0886261)` and `vardictcpp ae93fe39dc (Wolff 2026)`
Cloud Benchmark Host: `GCP n2-standard-32 (Intel Xeon Platinum 8373C Ice Lake, 32 vCPUs, 128 GiB RAM, 4x375 GB NVMe RAID0)` (`1.5` TB NVMe RAID0 scratch; exploratory staging also verified on `c3d-standard-32`)

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1. Supplementary Methods: Exact Command-Line Invocations Across All 60 Benchmark Cells

All `60` verification cells (`59` cloud compute cells plus `1` certified local-to-cloud evidence bundle `evidence_certified`) ran on `GCP n2-standard-32 (Intel Xeon Platinum 8373C Ice Lake, 32 vCPUs, 128 GiB RAM, 4x375 GB NVMe RAID0)` (`Ubuntu 24.04 LTS`, Linux kernel `6.8`, `1.5` TB 4-way NVMe RAID0 mounted at `/var/tmp/vardict-paper`). Timed runs used `/usr/bin/time -v` across `n = 3` independent technical replicates or `timeout 10800` (`3` hours, exit code `rc=124` on timeout; `rc=3` on JVM `OutOfMemoryError`).

1.0 Manuscript Figure and Table to 60-Cell Provenance Crosswalk

The crosswalk below maps each main-text Figure (`Figure 1` – `Figure 8`) and Table (`Table 1` – `Table 8`) to its benchmark directory in `benchmarks/gcp-study-v106/` (`Supplementary Table S1`):

- **`Figure 1` & `Table 1` (Architecture, Operational Modes, and 87-Arm Equivalence Sweep):** `s1_ceiling_amplicon`, `s1_ceiling_hg008`, `s1_gate7_ref_multiregion`, `s1_h1a_simple_tiles`, `s2_tierp_mode_parity_sweep`, `s3_rna_splice_parity_n3`, `s4_multimodal_and_cram_twin`, `evidence_certified`, and the `12` `sweep_*` cells (`sweep_4targets_full`, `sweep_R_chr1_full`, `sweep_hg002_chr20_exome`, `sweep_multimodal_and_cram_twin`, `sweep_rna_hg002_52gb`, `sweep_span_chr1_f0`, `sweep_tierp_13modes`, `sweep_tierp_SRR15006540`, `sweep_tierp_SRR8657348`, `sweep_tierp_somatic_76_86`, `sweep_ultradeep_mrd_stress`, `sweep_wgs_mrd_matrix`).

- **`Figure 2` & `Table 2` (Wolff et al. AML Cohort, Field Callers, and Contiguous `-R chr1`):** `s2_tierp_v2_SRR15006375_th1`, `s2_tierp_v2_SRR15006375_th8`, `s2_tierp_v2_SRR15006376_th1`, `s2_tierp_v2_SRR15006376_th8`, `s2_tierp_v2_SRR15006386_th1`, `s2_tierp_v2_SRR15006386_th8`, `s2_tierp_v2_SRR15006540_th1`, `s2_tierp_v2_SRR15006540_th8`, `s2_tierp_v2_SRR8657348_th1`, `s2_tierp_v2_SRR8657348_th8`, `s2_tierp_v2_somatic_76_86_th1`, `s2_tierp_v2_somatic_76_86_th8`, `s2_table3_R_chr1_java_200g_n3`, `s2_table3_R_chr1_rust_and_cpp`, and `s2_fig1_fig3_field_callers`.
- **`Figure 3` & `Table 3` (Germline/Somatic Truth Concordance, `QUAL` Operating Points, and Low-VAF Titration):** `s2_fig2_hg002_chr20_truth`, `s4_tier3_truth_and_qual_sweep`, `s5_grch38_field_callers_truth`, `s6_low_vaf_titration_truth_curve`, and `evidence_certified`.
- **`Figure 4` & `Table 4` (Multicore Scaling, Chromosome 1 Span-to-RSS Invariance, and Node Packing):** `s4_perf_budget_4targets_n3`, `s4_scale_exome_3way_th1`, `s4_scale_exome_3way_th2`, `s4_scale_exome_3way_th4`, `s4_scale_exome_3way_th8`, `s4_scale_exome_3way_th16`, `s4_scale_exome_3way_th32`, `s4_span_chr1_3way_th8`, `s4_span_chr1_3way_th16`, `s4_span_chr1_3way_th32`, `s4_span_chr1_f0`, and `s4_jobs_per_node_2x4c_n3`.
- **`Figure 5` & `Table 5` (Full-Genome WGS Unchunked Contigs vs. Pre-Chunked Callable Intervals):** `s5_wgs_hg002_full_contigs_3way_th32`, `s5_wgs_hg002_full_giab_3way_th32`, `s5_wgs_hg008_full_contigs_3way_th32`, and `s5_wgs_hg008_full_giab_3way_th32`.
- **`Figure 6` & `Table 6` (10-Chemistry Chr20 Panel, `-f 0.0` Noise Floor, and 70-Cell GA4GH `v3.5` Matrix):** `s5_cross_platform_short_read_chr20_panel` and `s6_ga4gh_v35_stratification_matrix`.
- **`Figure 7` & `Table 7` (Next-Generation `GIAB v5.0q T2T-Q100`, `CMRG v1.00`, `Mosaic v1.10`, and 11-Caller `HG008 v0.4` Benchmark):** `s6_hg002_v50q_cmrg_mosaic_eval` and `s6_hg008_modern_ml_and_commercial_callers`.
- **`Figure 8` & `Table 8` (Software Supply Chain and 10-Point `vardictcpp` Divergence Audit):** `s5_h4_maintenance_security_matrix` and `s2_tierp_mode_parity_sweep`.

1.1 Domain 1: Ceiling, Multi-Region Boundary, and Tile-Invariance Cells (`s1_*`)

Domain 1 comprises `4` cells (`s1_ceiling_amplicon`, `s1_ceiling_hg008`, `s1_gate7_ref_multiregion`, and `s1_h1a_simple_tiles`):

```
# Cell 1: s1_ceiling_amplicon (EGFR/MRE11a amplicon -a 10:0.95, 2 variants)
vardictrust -G hg19.fa -f 0.01 -N SAMPLE -b amplicon.bam \
  -a 10:0.95 --fisher -c 1 -S 2 -E 3 -g 4 --vcf amplicon.bed > rust_amp.vcf

# Cell 2: s1_ceiling_hg008 (NIST GIAB HG008 somatic slice, 18,426 variants)
vardictrust -G GRCh38_GIABv3.fasta -f 0.01 -N "HG008-T|HG008-N" \
  -b "HG008-T.bam|HG008-N.bam" -th 8 --fisher -z -c 1 -S 2 -E 3 -g 4 \
  --vcf hg008_slice.bed > rust_hg008.vcf

# Cell 3: s1_gate7_ref_multiregion (HG002 chr20 multi-region boundary gate, 2,572 variants)
vardictrust -G GRCh38.fa -f 0.01 -N HG002 -b hg002_chr20.bam \
  -th 8 --fisher -c 1 -S 2 -E 3 -g 4 --vcf chr20_multiregion.bed > rust_chr20_multiregion.vcf

# Cell 4: s1_h1a_simple_tiles (Tile-size invariance: 500,000, 1,000,000, 2,000,000, 10,000,000 bp -> 5,090 variants)
for TILE in 500000 1000000 2000000 10000000; do
  VARDICTRUST_TILE_BP="${TILE}" vardictrust -G GRCh38.fa -f 0.01 -N HCC1395 \
    -b HCC1395.bam -th 8 --fisher -c 1 -S 2 -E 3 -g 4 --vcf seqc2_slice.bed > "tile_${TILE}.vcf"
done
```

1.2 Domain 2: Wolff et al. AML Panel Replication, 13-Mode Sweep, Field Callers, and `-R chr1` (`s2_*`)

Domain 2 comprises `17` cells replicating and extending Wolff et al. (2026) [11]:

- **12 AML Panel Cells (`n = 3` technical replicates each at `-th 1` and `-th 8`):** `s2_tierp_v2_SRR15006375_th1`, `s2_tierp_v2_SRR15006375_th8`, `s2_tierp_v2_SRR15006376_th1`, `s2_tierp_v2_SRR15006376_th8`, `s2_tierp_v2_SRR15006386_th1`, `s2_tierp_v2_SRR15006386_th8`, `s2_tierp_v2_SRR15006540_th1`, `s2_tierp_v2_SRR15006540_th8`, `s2_tierp_v2_SRR8657348_th1`, `s2_tierp_v2_SRR8657348_th8`, `s2_tierp_v2_somatic_76_86_th1`, and `s2_tierp_v2_somatic_76_86_th8`.
- **Contiguous Chromosome 1 (`-R chr1:1-249250621`, `249,250,621` bp on `SRR8657348`):** `s2_table3_R_chr1_java_200g_n3` (`JAVA_OPTS="-Xmx4g"` OOM `rc=3` at `31.08` s; `JAVA_OPTS="-Xmx200g"` completes in `501.83` s at `106.47` GiB RSS) and `s2_table3_R_chr1_rust_and_cpp` (`VarDictRust` `9.75` s, `318.6` MB RSS vs `vardictcpp` `105.89` s, `10.58` GiB RSS).
- **13-Mode Functional Sweep & Field Callers:** `s2_tierp_mode_parity_sweep` (`13/13` modes), `s2_fig1_fig3_field_callers` (`FreeBayes`, `LoFreq`, `Mutect2`, `Octopus`, `Strelka2`), and `s2_fig2_hg002_chr20_truth` (`8,233` candidate variants).

```
# AML single-sample & paired somatic replication (s2_tierp_v2_*):
/usr/bin/time -v vardictrust -G hg19.fa -f 0.01 -N SRR8657348 \
  -b SRR8657348.bam -th 8 --fisher -z -c 1 -S 2 -E 3 -g 4 --vcf SRR8657348.bed > SRR8657348.vcf

# Table 3 contiguous chromosome 1 (-R chr1:1-249250621 on SRR8657348, s2_table3_R_chr1_*):
/usr/bin/time -v vardictrust -G hg19.fa -f 0.01 -N SRR8657348 \
  -b SRR8657348.bam -th 8 -R chr1:1-249250621 > rust_chr1.var
JAVA_OPTS="-Xmx200g" /usr/bin/time -v VarDict -G hg19.fa -f 0.01 -N SRR8657348 \
  -b SRR8657348.bam -th 8 -R chr1:1-249250621 > java_chr1.var
```

1.3 Domain 3: Full-Sample 51.8 GiB RNA-Seq Splice Calling (`s3_rna_splice_parity_n3`)

Cell `s3_rna_splice_parity_n3` (`n = 3` technical replicates) evaluates `-i` RNA splice-junction mode on the `51.8` GiB HG002 GM24385 RNA-seq alignment (`12,668` `.var` and `.splice.tsv` records, `4.94` s vs `23.31` s, `4.72` × speedup, `13.6` × RSS reduction):

```
/usr/bin/time -v vardictrust -G GRCh38.fa -f 0.01 -N HG002_RNA \
  -b HG002_GM24385_RNA.bam -i -th 8 -c 1 -S 2 -E 3 -g 4 gencode_exons.bed > rust_rna.var
```

1.4 Domain 4: Whole-Exome Scaling, Chromosome 1 Span-to-RSS, Node Packing, and `QUAL` ROC (`s4_*`)

Domain 4 comprises `15` cells (`s4_perf_budget_4targets_n3` , `s4_scale_exome_3way_th1` , `s4_scale_exome_3way_th2` , `s4_scale_exome_3way_th4` , `s4_scale_exome_3way_th8` , `s4_scale_exome_3way_th16` , `s4_scale_exome_3way_th32` , `s4_span_chr1_3way_th8` , `s4_span_chr1_3way_th16` , `s4_span_chr1_3way_th32` , `s4_span_chr1_f0` , `s4_jobs_per_node_2x4c_n3` , `s4_multimodal_and_cram_twin` , `s4_tier3_truth_and_qual_sweep` , and `evidence_certified`):

```
# Whole-exome scaling across 1, 2, 4, 8, 16, 32 threads on SEQC2 HCC1395 (s4_scale_exome_3way_th1..th32):
for TH in 1 2 4 8 16 32; do
  /usr/bin/time -v vardictrust -G GRCh38.fa -f 0.01 -N HCC1395 \
    -b HCC1395.bam -th "${TH}" --fisher -z -c 1 -S 2 -E 3 -g 4 \
    --vcf seqc2_exome.bed > "seqc2_th${TH}.vcf"
done

# Chromosome 1 span-to-RSS invariance across 1M, 10M, 100M, 249M bp on SEQC2 HCC1395 (s4_span_chr1_3way_th8..th32):
for SPAN in 1000000 10000000 100000000 248956422; do
  /usr/bin/time -v vardictrust -G GRCh38.fa -f 0.01 -N HCC1395 \
    -b HCC1395.bam -th 32 -R "chr1:1-${SPAN}" > "chr1_${SPAN}.var"
done

# All-positions (-f 0) pileup stress on chr1 10 Mbp (s4_span_chr1_f0) and 2x4c node packing
(s4_jobs_per_node_2x4c_n3):
/usr/bin/time -v vardictrust -G GRCh38.fa -f 0 -N HCC1395 -b HCC1395.bam -th 8 -R chr1:10000000-20000000 > chr1_f0.var
```

1.5 Multi-Regime Parameter Sweep: Comprehensive 87-Arm `.var` and `.vcf` Parity Sweep (`sweep_*`)

The `12` multi-regime sweep cells (`sweep_4targets_full` , `sweep_R_chr1_full` , `sweep_hg002_chr20_exome` , `sweep_multimodal_and_cram_twin` , `sweep_rna_hg002_52gb` , `sweep_span_chr1_f0` , `sweep_tierp_13modes` , `sweep_tierp_SRR15006540` , `sweep_tierp_SRR8657348` , `sweep_tierp_somatic_76_86` , `sweep_ultradeep_mrd_stress` , and `sweep_wgs_mrd_matrix`) validate `.var` and `.vcf` equivalence across `87` comparative arms (`87/87` `MATCH` , `100.00%` , `Supplementary Table S2`):

```
# Ultra-deep 45000x ctDNA MRD stress (-f 0.0005 and -f 0 with --fisher, sweep_ultradeep_mrd_stress):
vardictrust -G GRCh38.fa -f 0.0005 -N SAMPLE -b ultradeep_45000x.bam --fisher -c 1 -S 2 -E 3 -g 4 panel.bed >
rust_ultradeep_f0005.var
# WGS & exome MRD matrix (-f 0.001 and -f 0, sweep_wgs_mrd_matrix):
vardictrust -G GRCh38.fa -f 0.001 -N HG005 -b hg005_300x.bam --fisher -c 1 -S 2 -E 3 -g 4 --vcf mrd_targets.bed >
rust_hg005_300x_mrd.vcf
```

1.6 Domain 5: Full-Genome WGS (Unchunked Full Chromosomes vs. Pre-Chunked Callable Intervals), 10-Platform Chr20 Panel, and Supply Chain (`s5_*`)

Domain 5 comprises `7` cells (`s5_wgs_hg002_full_contigs_3way_th32` , `s5_wgs_hg002_full_giab_3way_th32` , `s5_wgs_hg008_full_contigs_3way_th32` , `s5_wgs_hg008_full_giab_3way_th32` , `s5_grch38_field_callers_truth` , `s5_cross_platform_short_read_chr20_panel` , and `s5_h4_maintenance_security_matrix`):

```
# Unchunked full chromosomes (chr1..chr22, s5_wgs_hg002_full_contigs_3way_th32 & s5_wgs_hg008_full_contigs_3way_th32):
timeout 10800 /usr/bin/time -v vardictrust -G GRCh38_GIABv3.fasta -f 0.01 -N HG002 \
  -b HG002_2x250_131GB.bam -th 32 --fisher -z -c 1 -S 2 -E 3 -g 4 --vcf grch38_22autosomes_full.bed >
hg002_unchunked.vcf

timeout 10800 /usr/bin/time -v vardictrust -G GRCh38_GIABv3.fasta -f 0.01 -N "HG008-T|HG008-N" \
  -b "HG008-T_195x.bam|HG008-N_169x.bam" -th 32 -U --fisher -z -c 1 -S 2 -E 3 -g 4 --vcf grch38_22autosomes_full.bed >
hg008_unchunked.vcf

# Pre-chunked <=50 kbp GIAB callable intervals (s5_wgs_hg002_full_giab_3way_th32 & s5_wgs_hg008_full_giab_3way_th32):
timeout 10800 /usr/bin/time -v vardictrust -G GRCh38_GIABv3.fasta -f 0.01 -N HG002 \
  -b HG002_2x250_131GB.bam -th 32 --fisher -z -c 1 -S 2 -E 3 -g 4 --vcf HG002_GRCh38_v421_50kb.bed >
hg002_giab50kb.vcf

timeout 10800 /usr/bin/time -v vardictrust -G GRCh38_GIABv3.fasta -f 0.01 -N "HG008-T|HG008-N" \
  -b "HG008-T_195x.bam|HG008-N_169x.bam" -th 32 -U --fisher -z -c 1 -S 2 -E 3 -g 4 --vcf HG008_GRCh38_v0.3_50kb.bed >
hg008_giab50kb.vcf
```

1.7 Domain 6: GA4GH v3.5 Stratification, GIAB v5.0q/CMRG/Mosaic, HG008 v0.4 11-Caller Matrix, and Low-VAF Titration (`s6_*`)

Domain 6 comprises `4` cells (`s6_ga4gh_v35_stratification_matrix` , `s6_hg002_v50q_cmrg_mosaic_eval` , `s6_hg008_modern_ml_and_commercial_callers` —including the official NIST GIAB `Liss_lab` somatic callsets for `DRAGEN v4.2.4` , `DeepSomatic v1.6.0` , and `ClairS v0.4.1` , alongside `Lancet2 (NYGC)` , `Hartwig SAGE` , and `Ultima DeepVariant Somatic` —and `s6_low_vaf_titration_truth_curve`):

```
# Controlled in silico tumor-purity titration (20%, 10%, 5%, 2%, 1% purity at -f 0.002,
s6_low_vaf_titration_truth_curve):
for PURITY in 20 10 5 2 1; do
  vardictrust -G GRCh38_GIABv3.fasta -f 0.002 -N "HG008_P${PURITY}|HG008-N" \
    -b "hg008_titration_p${PURITY}.bam|HG008-N.bam" -th 16 --fisher -z -c 1 -S 2 -E 3 -g 4 \
    --vcf hg008_slice.bed > "rust_titration_p${PURITY}.vcf"
done

# Next-gen GIAB v5.0q T2T-Q100, CMRG v1.00, and Mosaic v1.10 evaluation (s6_hg002_v50q_cmrg_mosaic_eval):
rtg vcfeval -b HG002_GRCh38_v5.0q_T2T.vcf.gz -e HG002_GRCh38_v5.0q_T2T.bed \
  -c hg002_unchunked.vcf.gz -t GRCh38.sdf -o eval_v50q_unchunked
```

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2. Supplementary Note 1: Analytical Derivation of JVM (`VarDictJava`) and `vardictcpp` Heap Retention (Unchunked Chromosomes vs. Pre-Chunked Callable Intervals)

In `VarDictJava` v1.8.3 (`CigarParser.java` and `VariationRealigner.java`), the pileup engine processes each BED region *[S, E]* of length $L = E - S + 1$ by instantiating two region-wide heap structures before emitting any variant row:

1. A reference base map `Reference.referenceSequences` (`Map<Integer, Character>`) holding *O(L)* boxed `java.lang.Integer` (`16` bytes) and `java.lang.Character` (`16` bytes) objects plus `HashMap$Node` headers (`32` bytes), consuming ≈ *64–96* bytes per reference base (≈ *16–24* GiB for human `chr1` of length $L = 249{,}250{,}621$ bp per worker thread).
2. A position-indexed variation map `Map<Integer, VariationMap<String, Variation>> nonInsertionVariants` and `insertionVariants` , where each covered coordinate $p \in [S, E]$ allocates a `VariationMap` and one `Variation` object per observed allele. At sequencing depth D, the active heap footprint across T worker threads scales as:

$$\mathrm{RSS}_{\text{Java / C++}}(L, T) = \Theta\left(\Sigma_{t=1}^{T} L_t \cdot (c_{\text{ref}} + c_{\text{var}} \cdot \bar{a}(D))\right)$$

where L_t is the region length assigned to thread t and $\bar{a}(D)$ is the mean allele count per locus. When $L_t \le 50$ kbp (pre-chunked intervals), garbage collection reclaims objects between intervals (`34.14` GB on HG002; `62.51` GB with timeout `rc=124` on HG008). When $L_t \in [50, 249]$ Mbp (unchunked chromosomes or `-R chr1:1-249250621`), live objects remain reachable until the chromosome finishes, driving `VarDictJava` to `106.47` GiB on `chr1` and `>213.7` GB (`rc=3` `OutOfMemoryError`) on multi-threaded full genomes (`Table 2` , `Table 5`). `vardictcpp` similarly retains per-region C++ hash tables across each interval, consuming `10.58` GiB on `chr1` , `31.91` GB on HG002 unchunked chromosomes, and `91.85` GB (`rc=124` timeout) on HG008 unchunked chromosomes unless an external script pre-splits the BED file (`CPP_CHUNK=50,000`).

`VarDictRust` leaves intervals ≤ *1,000,000* bp untiled and partitions any region $L > 1{,}000{,}000$ bp into tiles of default width $W = 100$ kbp with halo $H = 10$ kbp (`200` kbp in `-i` splice mode; `Figure 1b`), bounding memory to:

$$\mathrm{RSS}_{\mathrm{VarDictRust}}(L, T) = O(T \cdot (W + 2H)) \perp L$$

Increasing L by `250` × on chromosome 1 (`1` Mbp to `249` Mbp, `s4_span_chr1_3way_th8` .. `th32`) changes `VarDictRust` RSS by `1.18` ×– `1.41` × (`1.38` – `3.69` GiB), allowing unchunked full-genome calling in `4.93` GB (HG002) and `12.14` GB (HG008).

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3. Supplementary Note 2: Ten-Point Technical Audit of `vardictcpp` Divergences (`CPP-1` through `CPP-10`)

Across cells `s2_tierp_mode_parity_sweep` , `s5_h4_maintenance_security_matrix` , `s5_wgs_hg002_full_contigs_3way_th32` , and `s6_low_vaf_titration_truth_curve` , we cataloged ten reproducibility and capability gaps in `vardictcpp ae93fe39dc (Wolff 2026)` (`Table 8` , `Supplementary Table S6`):

- **`CPP-1` (VCF Header & `INFO/END` Schema):** `vardictcpp` omits the `##INFO=<ID=END>` header definition and diverges from `var2vcf_valid.pl` / `var2vcf_paired.pl` VCF 4.3 schema formatting.
- **`CPP-2` (CRISPR (`-J` / `-j`) & Structural Variant Modes):** `vardictcpp` sets `--nosv` (`-U`) by default and omits CRISPR cleavage-site (`-J` / `-j`) and structural variant (`StructuralVariantsProcessor`) assembly (`1/13` modes failed in `s2_tierp_mode_parity_sweep` , `12/13` exact).
- **`CPP-3` (Amplicon (`-a`) Output Format):** In amplicon primer-clipping mode (`-a 10:0.95` , `s1_ceiling_amplicon`), `vardictcpp` emits raw tab-separated `.var` text instead of VCF (`0` VCF lines), breaking downstream `bcftools` and `rtg vcfeval` consumers.
- **`CPP-4` (Concatenated CLI Flag Parsing):** Passing bundled single-character flags (e.g., `-z -c 1` combined or legacy `VarDictJava` bundled options) causes `vardictcpp` to fail CLI parsing with exit code `120` .
- **`CPP-5` (50 kbp Chunk Boundary Deduplication):** Without external BED splitting (`--chunk 50,000` / `CPP_CHUNK=50,000`), `vardictcpp` consumes `10.58` GiB on `chr1` , `31.91` GB on HG002 unchunked chromosomes, and `91.85` GB (`rc=124` timeout) on HG008 unchunked chromosomes. When intervals are chunked, `vardictcpp` lacks boundary deduplication, emitting `+309,569` duplicate/shifted calls on HG002 unchunked chromosomes (`9,953,338` vs `9,643,769`) and `+394` on pre-chunked intervals (`8,111,565` vs `8,111,171`).
- **`CPP-6` (Missing `##contig` Header Lines):** `vardictcpp` omits `##contig=<ID=...,length=...>` lines from VCF headers (`VarDictJava` issue `#320`), causing strict VCF validators and `rtg vcfeval` header checks to warn or fail.
- **`CPP-7` (GRCh37 / b37 Chromosome Naming):** `vardictcpp` hardcodes `chr` -prefix assumptions in internal chromosome handling, breaking or diverging on `b37` / `GRCh37` numeric contigs (`1` .. `22` , `X` , `Y` , `MT`).
- **`CPP-8` (BAM/CRAM Reference & Decode):** `vardictcpp` diverges on CRAM reference decoding (`-G ref.fa`) and requires dynamic linkage against external `libhts` , `libcurl` , `liblzma` , `libbz2` , and `libdeflate` , whereas `VarDictRust` natively decodes BAM and CRAM in a `4.56` MB zero-dependency static `musl` binary (`s4_multimodal_and_cram_twin`).
- **`CPP-9` (MNP Phasing, Tie-Break & Somatic SV Drops):** `vardictcpp` diverges from `VarDictJava` on complex MNP phasing, realignment tie-breaking, and somatic variant retention. At `20%` tumor purity (`s6_low_vaf_titration_truth_curve`), `vardictcpp` drops somatic indel F_1 to `0.2466` (recall `15.65%`) and SNV F_1 to `0.6914` (recall `57.07%`), whereas `VarDictRust` matches `VarDictJava` (`Indel` $F_1 = 0.3104$, recall `20.75%` ; `SNV` $F_1 = 0.7512$, recall `65.59%`).
- **`CPP-10` (RNA-Seq Splice Mode (`-i` .var / `.splice.tsv`):** `vardictcpp` does not implement `-i` exon-junction splice-site accounting (`s3_rna_splice_parity_n3` , `51.8` GiB RNA-seq BAM, `12,668` records).

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4. Supplementary Note 3: Biostatistical Methodology, Exact Clopper-Pearson Intervals, and `--fix` Governance

4.1 Exact Clopper-Pearson Binomial Confidence Intervals

For binary oracle-concordance and completion rates (k successes out of n independent cells/arms), we compute exact two-sided $1 - \alpha = 95\%$ Clopper-Pearson binomial CIs $[p_{\mathrm{lo}}, p_{\mathrm{hi}}]$ via the inverse Beta distribution:

$$p_{\mathrm{lo}} = \mathrm{Beta}^{-1}(\alpha/2;\ k,\ n - k + 1), \qquad p_{\mathrm{hi}} = \mathrm{Beta}^{-1}(1 - \alpha/2;\ k + 1,\ n - k)$$

with $p_{\mathrm{lo}} = 0$ when $k = 0$ and $p_{\mathrm{hi}} = 1$ when $k = n$:

- Complete `60`-cell verification campaign (`60/60` exit `0`): `100%` (`95%` CI `94.04%–100.00%`)
- `13`-mode functional parity sweep (`13/13`): `100%` (`95%` CI `75.29%–100.00%`)
- Extended coverage functional modalities (`17/17`): `100%` (`95%` CI `80.49%–100.00%`)
- Multimodal clinical and edge-case slices (`18/18`): `100%` (`95%` CI `81.47%–100.00%`)
- Cross-platform chr20 runs where `VarDictJava` completed (`11/11`): `100%` (`95%` CI `71.51%–100.00%`)
- Cross-platform `-f 0` somatic noise-floor `STRICT-BYTES` sweep (`18/18`): `100%` (`95%` CI `81.47%–100.00%`)
- GA4GH `v3.5` `70`-cell stratification matrix (`70/70`): `100%` (`95%` CI `94.87%–100.00%`)
- Multi-regime parity sweep across `12` suites (`87/87`): `100.00%` (`95%` CI `95.85%–100.00%` ; includes untiled ≤ *1,000,000* bp BED suites and single-window `--tile-size` runs on `sweep_span_chr1_f0` [`7,077` `.var` / `7,032` `.vcf`] and `sweep_R_chr1_full` [`4,705` `.var`]; `Supplementary Table S2`)
- NIST Mosaic SNV `v1.10` sensitivity (`84/85`): `98.82%` (`95%` CI `93.62%–99.97%`)

4.2 Non-Parametric Runtime and Memory Hypothesis Testing

Across the *N = 12* AML sample-by-thread cells (`s2_tierp_v2_*` , `n = 3` technical replicates each), we report the mean and replicate range (`min–max` , `Supplementary Table S3`). Comparing `VarDictRust` against `VarDictJava` across *N = 12* yields two-sided Mann-Whitney *U = 4.0* (*p = 9.735e-05*) for wall time, *U = 0.0* (*p = 3.658e-05*) for peak RSS, and an exact paired sign-flip permutation p-value of $p = 2^{-12} = 0.00024$.

4.3 Release Governance: `v1.0.x` Parity Versus `v1.1+` Opt-In `--fix` Flags

All `v1.0.x` releases (`v1.0.6`) enforce strict oracle parity on valid inputs. Output-changing upstream fixes (`#92` `VD > DP` , `#62` over-filtering, `#292` `SHIFT3` , `#342` `<dup-N>` symbolic alleles) are gated behind opt-in `--fix <id>` flags starting in `v1.1.0` , while parity-restoring fixes (`splice-tile-ownership` , `mate-window-ownership`) ship as verified defaults.

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5. Supplementary Tables (`Supplementary Table S1` – `Supplementary Table S6`) and Complete 60-Cell Verification Ledger

Supplementary Table S1. Complete 60-Experiment Benchmark Provenance Inventory, Execution Status & Manuscript Crosswalk

Archive Directory ID	Evaluation Suite	Assay Class[a]	Exit Code[b]	Artifacts	Manuscript Targets	Parity / Execution Verdict	Experiment Description
s1_ceiling_amplicon	Baseline Equivalence & Tiling	Targeted Panel	0	6	Figure 1, Figure 4, Table 1, Supplementary Table S1	100% Exact Match (2 variants)	EGFR/MRE11a amplicon (-a) 3-way ceiling & exact parity
s1_ceiling_hg008	Baseline Equivalence & Tiling	Whole Genome	0	6	Figure 1, Figure 4, Table 1, Supplementary Table S1	100% Exact Match (18,426 variants)	NIST GIAB HG008 somatic slice 3-way ceiling & exact parity
s1_gate7_ref_multiregion	Baseline Equivalence & Tiling	Whole Genome	0	5	Figure 1, Figure 2, Table 1, Supplementary Table S1	100% Exact Match (2,572 variants)	HG002 chr20 multi-region reference boundary parity gate
s1_h1a_simple_tiles	Baseline Equivalence & Tiling	Whole Exome	0	5	Figure 1, Figure 2, Table 1, Supplementary Table S1	100% Exact Match across 4 tile sizes (5,090 variants)	Tile-size invariance (0.5, 1, 2, 10 Mbp) on SEQC2 HCC1395
s2_fig1_fig3_field_callers	AML Panel & Field-Caller Replication [11]	Targeted Panel	0	73	Figure 1, Figure 2, Table 2, Supplementary Table S1	16/16 field-caller runs completed (VarDictRust == VarDictJava)	Wolff et al. [11] Fig 1 & Fig 3 field callers (FreeBayes, LoFreq, Mutect2, Octopus, Strelka2) at 1c/8c
s2_fig2_hg002_chr20_truth	AML Panel & Field-Caller Replication [11]	Whole Genome	0	20	Figure 1, Figure 3, Table 2, Table 3, Supplementary Table S1	100% Exact Match (7,422 variants; 7 callers evaluated)	Wolff et al. [11] Fig 2 / Daniels et al. [20] HG002 chr20 exome 7-caller truth accuracy
s2_table3_R_chr1_java_200g_n3	AML Panel & Field-Caller Replication [11]	Targeted Panel	0	6	Figure 1, Figure 4, Table 2, Table 4, Supplementary Table S1	VarDictJava -Xmx4g OOM (rc=3); completes at -Xmx200g (n=3)	Wolff et al. [11] Table 3 -R chr1:1-249250621 Java -Xmx4g OOM & -Xmx200g completion (n=3)

Archive Directory ID	Evaluation Suite	Assay Class[a]	Exit Code[b]	Artifacts	Manuscript Targets	Parity / Execution Verdict	Experiment Description
s2_table3_R_chr1_rust_and_cpp	AML Panel & Field-Caller Replication [11]	Targeted Panel	0	10	Figure 1, Figure 4, Table 2, Table 4, Supplementary Table S1	VarDictRust & vardictcpp complete in bounded memory (n=3)	Wolff et al. [11] Table 3 -R chr1:1-249250621 VarDictRust vs vardictcpp (n=3)
s2_tierp_mode_parity_sweep	AML Panel & Field-Caller Replication [11]	Targeted Panel	0	14	Figure 1, Figure 2, Figure 8, Table 1, Table 2, Table 8, Supplementary Table S1, Supplementary Table S6	100% Exact Match across all 13 functional CLI modes	13-mode 3-way functional parity sweep (Simple, Somatic, Amplicon, Splice, SV, CRAM, Fisher, -f 0)
s2_tierp_v2_SRR15006375_th1	AML Panel & Field-Caller Replication [11]	Targeted Panel	0	19	Figure 1, Figure 2, Table 2, Supplementary Table S1, Supplementary Table S3	100% Exact Match (4,056 .var rows; 3,966 VCF variants)	Clinical AML AML panel SRR15006375 3-way parity & timing at 1 thread (n=3)
s2_tierp_v2_SRR15006375_th8	AML Panel & Field-Caller Replication [11]	Targeted Panel	0	19	Figure 1, Figure 2, Table 2, Supplementary Table S1, Supplementary Table S3	100% Exact Match (4,056 .var rows; 3,966 VCF variants)	Clinical AML AML panel SRR15006375 3-way parity & timing at 8 threads (n=3)
s2_tierp_v2_SRR15006376_th1	AML Panel & Field-Caller Replication [11]	Targeted Panel	0	19	Figure 1, Figure 2, Table 2, Supplementary Table S1, Supplementary Table S3	100% Exact Match (12,739 .var rows; 12,655 VCF variants)	Clinical AML AML panel SRR15006376 3-way parity & timing at 1 thread (n=3)
s2_tierp_v2_SRR15006376_th8	AML Panel & Field-Caller Replication [11]	Targeted Panel	0	19	Figure 1, Figure 2, Table 2, Supplementary Table S1, Supplementary Table S3	100% Exact Match (12,739 .var rows; 12,655 VCF variants)	Clinical AML AML panel SRR15006376 3-way parity & timing at 8 threads (n=3)
s2_tierp_v2_SRR15006386_th1	AML Panel & Field-Caller Replication [11]	Targeted Panel	0	19	Figure 1, Figure 2, Table 2, Supplementary Table S1, Supplementary Table S3	100% Exact Match (4,014 .var rows; 3,925 VCF variants)	Clinical AML AML panel SRR15006386 3-way parity & timing at 1 thread (n=3)
s2_tierp_v2_SRR15006386_th8	AML Panel & Field-Caller Replication [11]	Targeted Panel	0	19	Figure 1, Figure 2, Table 2, Supplementary Table S1, Supplementary Table S3	100% Exact Match (4,014 .var rows; 3,925 VCF variants)	Clinical AML AML panel SRR15006386 3-way parity & timing at 8 threads (n=3)
s2_tierp_v2_SRR15006540_th1	AML Panel & Field-Caller Replication [11]	Targeted Panel	0	19	Figure 1, Figure 2, Table 2, Supplementary Table S1, Supplementary Table S3	100% Exact Match (16,031 .var rows; 15,912 VCF variants)	Clinical AML AML panel SRR15006540 3-way parity & timing at 1 thread (n=3)
s2_tierp_v2_SRR15006540_th8	AML Panel & Field-Caller Replication [11]	Targeted Panel	0	19	Figure 1, Figure 2, Table 2, Supplementary Table S1, Supplementary Table S3	100% Exact Match (16,031 .var rows; 15,912 VCF variants)	Clinical AML AML panel SRR15006540 3-way parity & timing at 8 threads (n=3)
s2_tierp_v2_SRR8657348_th1	AML Panel & Field-Caller Replication [11]	Targeted Panel	0	19	Figure 1, Figure 2, Table 2, Supplementary Table S1, Supplementary Table S3	100% Exact Match (54,448 .var rows; 53,224 VCF variants)	Clinical AML AML panel SRR8657348 (773,667 intervals) 3-way parity & timing at 1 thread (n=3)
s2_tierp_v2_SRR8657348_th8	AML Panel & Field-Caller Replication [11]	Targeted Panel	0	19	Figure 1, Figure 2, Table 2, Supplementary Table S1, Supplementary Table S3	100% Exact Match (54,448 .var rows; 53,224 VCF variants)	Clinical AML AML panel SRR8657348 (773,667 intervals) 3-way parity & timing at 8 threads (n=3)
s2_tierp_v2_somatic_76_86_th1	AML Panel & Field-Caller Replication [11]	Targeted Panel	0	17	Figure 1, Figure 2, Table 2, Supplementary Table S1, Supplementary Table S3	100% Exact Match (14,126 .var rows; 14,000 VCF variants)	Clinical AML somatic pair SRR15006376 / SRR15006386 3-way parity & timing at 1 thread (n=3)
s2_tierp_v2_somatic_76_86_th8	AML Panel & Field-Caller Replication [11]	Targeted Panel	0	17	Figure 1, Figure 2, Table 2, Supplementary Table S1, Supplementary Table S3	100% Exact Match (14,126 .var rows; 14,000 VCF variants)	Clinical AML somatic pair SRR15006376 / SRR15006386 3-way parity & timing at 8 threads (n=3)
s3_rna_splice_parity_n3	Transcriptome RNA-Seq Splice Parity	Transcriptome RNA-Seq	0	10	Figure 1, Figure 4, Table 1, Table 4, Supplementary Table S1	100% Exact Match (12,668 rows)	Full-sample 51.8 GiB HG002 GM24385 RNA-seq (-i splice mode) parity & timing (n=3)

Archive Directory ID	Evaluation Suite	Assay Class[a]	Exit Code[b]	Artifacts	Manuscript Targets	Parity / Execution Verdict	Experiment Description
s4_jobs_per_node_2x4c_n3	Multi-Thread Scaling, Span & ROC	Whole Exome	0	7	Figure 1, Figure 4, Table 4, Supplementary Table S1	2x4-thread node packing completed (0.859 GiB combined RSS, n=3)	Concurrent 2x4-core node-packing throughput & RSS on SEQC2 HCC1395 exome (n=3)
s4_multimodal_and_cram_twin	Multi-Thread Scaling, Span & ROC	Whole Genome	0	3	Figure 1, Figure 2, Table 1, Supplementary Table S1	100% Exact Match (0 textual differences within eps=1e-6)	Lossless BAM-vs-CRAM twin zero-diff check & version verification
s4_perf_budget_4targets_n3	Multi-Thread Scaling, Span & ROC	Panel, Exome & WGS	0	54	Figure 1, Figure 4, Table 4, Supplementary Table S1	3-way timing & RSS verified across all 4 benchmark targets (n=3)	3-way performance budget across amplicon, HG002 germline, SEQC2 somatic exome, and HG008 somatic (n=3)
s4_scale_exome_3way_th1	Multi-Thread Scaling, Span & ROC	Whole Exome	0	11	Figure 1, Figure 4, Figure 8, Table 4, Supplementary Table S1, Supplementary Table S3	100% Exact Match (339,698 variants)	SEQC2 HCC1395 whole-exome 3-way scaling at 1 thread (n=3)
s4_scale_exome_3way_th2	Multi-Thread Scaling, Span & ROC	Whole Exome	0	12	Figure 1, Figure 4, Figure 8, Table 4, Supplementary Table S1, Supplementary Table S3	100% Exact Match (339,698 variants)	SEQC2 HCC1395 whole-exome 3-way scaling at 2 threads (n=3)
s4_scale_exome_3way_th4	Multi-Thread Scaling, Span & ROC	Whole Exome	0	12	Figure 1, Figure 4, Figure 8, Table 4, Supplementary Table S1, Supplementary Table S3	100% Exact Match (339,698 variants)	SEQC2 HCC1395 whole-exome 3-way scaling at 4 threads (n=3)
s4_scale_exome_3way_th8	Multi-Thread Scaling, Span & ROC	Whole Exome	0	12	Figure 1, Figure 4, Figure 8, Table 4, Supplementary Table S1, Supplementary Table S3	100% Exact Match (339,698 variants)	SEQC2 HCC1395 whole-exome 3-way scaling at 8 threads (n=3)
s4_scale_exome_3way_th16	Multi-Thread Scaling, Span & ROC	Whole Exome	0	12	Figure 1, Figure 4, Figure 8, Table 4, Supplementary Table S1, Supplementary Table S3	100% Exact Match (339,698 variants)	SEQC2 HCC1395 whole-exome 3-way scaling at 16 threads (n=3)
s4_scale_exome_3way_th32	Multi-Thread Scaling, Span & ROC	Whole Exome	0	12	Figure 1, Figure 4, Figure 8, Table 4, Supplementary Table S1, Supplementary Table S3	100% Exact Match (339,698 variants)	SEQC2 HCC1395 whole-exome 3-way scaling at 32 threads (n=3)
s4_span_chr1_3way_th8	Multi-Thread Scaling, Span & ROC	Whole Exome	0	37	Figure 1, Figure 4, Table 4, Supplementary Table S1, Supplementary Table S3	Bounded memory verified across 1–249 Mbp spans (<1.50x RSS ratio, n=3)	Chromosome 1 span-to-RSS scaling (1, 10, 100, 249 Mbp) at 8 threads (n=3)
s4_span_chr1_3way_th16	Multi-Thread Scaling, Span & ROC	Whole Exome	0	37	Figure 1, Figure 4, Table 4, Supplementary Table S1, Supplementary Table S3	Bounded memory verified across 1–249 Mbp spans (<1.50x RSS ratio, n=3)	Chromosome 1 span-to-RSS scaling (1, 10, 100, 249 Mbp) at 16 threads (n=3)
s4_span_chr1_3way_th32	Multi-Thread Scaling, Span & ROC	Whole Exome	0	37	Figure 1, Figure 4, Table 4, Supplementary Table S1, Supplementary Table S3	Bounded memory verified across 1–249 Mbp spans (<1.50x RSS ratio, n=3)	Chromosome 1 span-to-RSS scaling (1, 10, 100, 249 Mbp) at 32 threads (n=3)
s4_span_chr1_f0	Multi-Thread Scaling, Span & ROC	Whole Exome	0	9	Figure 1, Figure 4, Table 1, Table 4, Supplementary Table S1	100% Exact Match (7,032 variants)	Chromosome 1 10 Mbp all-positions (-f 0) parity & memory stress (n=3)
s4_tier3_truth_and_qual_sweep	Multi-Thread Scaling, Span & ROC	Exome & WGS	0	24	Figure 1, Figure 3, Figure 7, Table 3, Table 7, Supplementary Table S1, Supplementary Table S4	100% Exact Match across HG002, SEQC2 & HG008 QUAL ROC sweeps	rtg vcfeval truth concordance & QUAL ROC sweep (0..200) on HG002, SEQC2, HG008
evidence_certified	Multi-Thread Scaling, Span & ROC	Panel, Exome & WGS	0	16	Figure 1, Figure 2, Figure 3, Table 1, Table 3, Supplementary Table S1	100% Exact Match (17/17 extended modes, 18/18 slices, HG005 & NA24385)	GCP-certified evidence bundle (17-modality extended coverage, 18-slice multimodal, HG005, NA24385-b37)

Archive Directory ID	Evaluation Suite	Assay Class[a]	Exit Code[b]	Artifacts	Manuscript Targets	Parity / Execution Verdict	Experiment Description
sweep_4targets_full	Full-Core Multi-Regime Parity Sweep	Panel, Exome & WGS	0	15	Figure 1, Figure 2, Table 1, Supplementary Table S1, Supplementary Table S2	100% Exact Match (4/4 sub-arms)	Full-core multi-regime .var & .vcf parity sweep across amplicon, hg002, seqc2, hg008
sweep_R_chr1_full	Full-Core Multi-Regime Parity Sweep	Targeted Panel	0	6	Figure 1, Figure 2, Table 1, Supplementary Table S1, Supplementary Table S2	100% Exact Match (1/1 sub-arm)	Full-core multi-regime .var parity sweep on -R chr1:1-249250621 (249 Mbp single region)
sweep_hg002_chr20_exome	Full-Core Multi-Regime Parity Sweep	Whole Genome	0	9	Figure 1, Figure 2, Table 1, Supplementary Table S1, Supplementary Table S2	100% Exact Match (2/2 sub-arms)	Full-core multi-regime .var & .vcf parity sweep on HG002 chr20 exome (Daniels et al. [20])
sweep_multimodal_and_cram_twin	Full-Core Multi-Regime Parity Sweep	Whole Genome	0	11	Figure 1, Figure 2, Table 1, Supplementary Table S1, Supplementary Table S2	100% Exact Match (4/4 sub-arms)	Full-core multi-regime multimodal slices, extended coverage, and HG008 CRAM twin timing & parity
sweep_rna_hg002_52gb	Full-Core Multi-Regime Parity Sweep	Transcriptome RNA-Seq	0	10	Figure 1, Figure 2, Table 1, Supplementary Table S1, Supplementary Table S2	100% Exact Match (2/2 sub-arms)	Full-core multi-regime .var & .splice.tsv parity sweep on 51.8 GiB HG002 RNA-seq
sweep_span_chr1_f0	Full-Core Multi-Regime Parity Sweep	Whole Exome	0	9	Figure 1, Figure 2, Table 1, Supplementary Table S1, Supplementary Table S2	100% Exact Match (2/2 sub-arms)	Full-core multi-regime .var & .vcf parity sweep on chr1 10 Mbp at -f 0
sweep_tierp_13modes	Full-Core Multi-Regime Parity Sweep	Targeted Panel	0	49	Figure 1, Figure 2, Table 1, Supplementary Table S1, Supplementary Table S2	100% Exact Match (12/12 sub-arms)	Full-core multi-regime .var parity sweep across 12 Clinical AML operational mode sub-arms
sweep_tierp_SRR15006540	Full-Core Multi-Regime Parity Sweep	Targeted Panel	0	9	Figure 1, Figure 2, Table 1, Supplementary Table S1, Supplementary Table S2	100% Exact Match (2/2 sub-arms)	Full-core multi-regime .var & .vcf parity sweep on Clinical AML SRR15006540
sweep_tierp_SRR8657348	Full-Core Multi-Regime Parity Sweep	Targeted Panel	0	9	Figure 1, Figure 2, Table 1, Supplementary Table S1, Supplementary Table S2	100% Exact Match (2/2 sub-arms)	Full-core multi-regime .var & .vcf parity sweep on Clinical AML SRR8657348
sweep_tierp_somatic_76_86	Full-Core Multi-Regime Parity Sweep	Targeted Panel	0	9	Figure 1, Figure 2, Table 1, Supplementary Table S1, Supplementary Table S2	100% Exact Match (2/2 sub-arms)	Full-core multi-regime .var & .vcf parity sweep on Clinical AML somatic pair SRR15006376 / SRR15006386
sweep_ultradeep_mrd_stress	Full-Core Multi-Regime Parity Sweep	Panel & WGS	0	107	Figure 1, Figure 2, Table 1, Supplementary Table S1, Supplementary Table S2	100% Exact Match (30/30 sub-arms)	Full-core multi-regime ultra-deep MRD stress test (-f 0.0005 and -f 0 with --fisher up to 45,000x)
sweep_wgs_mrd_matrix	Full-Core Multi-Regime Parity Sweep	Exome & WGS	0	86	Figure 1, Figure 2, Table 1, Supplementary Table S1, Supplementary Table S2	100% Exact Match (24/24 sub-arms)	Full-core multi-regime WGS & exome MRD matrix (-f 0.001 and -f 0 across HG002, HG005 300x, SEQC2, HG008)
s5_cross_platform_short_read_chr20_panel	Full-Genome WGS, 10-Platform & Security	10-Platform Chr20 WGS	0	420	Figure 1, Figure 6, Table 6, Supplementary Table S1, Supplementary Table S4	100% Exact Match (11/11 completing Chr20 runs & 18/18 noise-floor runs)	10-platform sequencing chemistry chr20 panel (Part A -f 0.01/-f 0 std/UG-HCR; Part B 18-arm noise floor; Ultima CRAM twin)
s5_grch38_field_callers_truth	Full-Genome WGS, 10-Platform & Security	Whole Genome	0	78	Figure 1, Figure 3, Table 3, Supplementary Table S1	19/19 caller-cohort evaluations completed (VarDictRust == VarDictJava)	Native GRCh38 germline (HG002 chr20) & somatic (HG008 slice, SEQC2) field-caller truth evaluation
s5_h4_maintenance_security_matrix	Full-Genome WGS, 10-Platform & Security	Cross-Modal Security & Parity	0	13	Figure 1, Figure 8, Table 8, Supplementary Table S1, Supplementary Table S6	Static musl equivalence verified & CPP-1..10 cataloged	supply-chain security, static musl binary, CLI flag parity, and CPP-1..10 divergence matrix

Archive Directory ID	Evaluation Suite	Assay Class[a]	Exit Code[b]	Artifacts	Manuscript Targets	Parity / Execution Verdict	Experiment Description
s5_wgs_hg002_full_contigs_3way_th32	Full-Genome WGS, 10-Platform & Security	Full-Genome WGS (chr1–chr22)	0	14	Figure 1, Figure 5, Table 5, Supplementary Table S1	VarDictJava OOM (rc=3); VarDictRust completes (9,643,769 variants)	Full-genome HG002 (130.8 GB 2x250 bp) (22 unchunked chromosomes) 3-way scaling
s5_wgs_hg002_full_giab_3way_th32	Full-Genome WGS, 10-Platform & Security	Full-Genome WGS (chr1–chr22)	0	27	Figure 1, Figure 5, Table 1, Table 5, Supplementary Table S1	100% Exact Match (8,111,171 variants)	Full-genome HG002 (130.8 GB 2x250 bp) (≤50 kbp GIAB callable intervals) 3-way parity & truth
s5_wgs_hg008_full_contigs_3way_th32	Full-Genome WGS, 10-Platform & Security	Full-Genome WGS (chr1–chr22)	0	12	Figure 1, Figure 5, Table 5, Supplementary Table S1	VarDictJava OOM (rc=3) & C++ timeout; VarDictRust completes (13,067,287 variants)	Full-genome HG008 (329.0 GB 364x T/N) (22 unchunked chromosomes) 3-way scaling
s5_wgs_hg008_full_giab_3way_th32	Full-Genome WGS, 10-Platform & Security	Full-Genome WGS (chr1–chr22)	0	20	Figure 1, Figure 5, Table 5, Supplementary Table S1	VarDictJava & C++ timeout (rc=124); VarDictRust completes (10,195,118 variants)	Full-genome HG008 (329.0 GB 364x T/N) (≤50 kbp GIAB v0.4 callable intervals) 3-way scaling & v0.3/v0.4 truth
s6_ga4gh_v35_stratification_matrix	Next-Gen GIAB, GA4GH v3.5 & Low-VAF Titration	GA4GH v3.5 Stratification [17]	0	2	Figure 1, Figure 6, Table 6, Supplementary Table S1, Supplementary Table S5	100% Exact Match across all 70 GA4GH v3.5 stratification cells	70-cell GA4GH v3.5 genomic stratification matrix (7 strata x 10 sequencing platforms)
s6_hg002_v50q_cmrg_mosaic_eval	Next-Gen GIAB, GA4GH v3.5 & Low-VAF Titration	Next-Gen GIAB Truth [14–16, 19]	0	2	Figure 1, Figure 7, Table 7, Supplementary Table S1	12/12 evaluations completed (100% Match on callable BED)	HG002 GIAB v5.0q T2T-Q100, CMRG v1.00, and Mosaic SNV v1.10 evaluation
s6_hg008_modern_ml_and_commercial_callers	Next-Gen GIAB, GA4GH v3.5 & Low-VAF Titration	GIAB HG008 v0.4 Somatic [19]	0	2	Figure 1, Figure 7, Table 7, Supplementary Table S1	27/27 caller-benchmark evaluations completed across 11 callers	HG008 GIAB v0.4 benchmark across VarDictRust, 7 NIST Liss_lab callers (DRAGEN, DeepSomatic, ClairS, Lancet2, SAGE, Ultima DV), Mutect2, Strelka2, LoFreq
s6_low_vaf_titration_truth_curve	Next-Gen GIAB, GA4GH v3.5 & Low-VAF Titration	Somatic Low-VAF Titration [19]	0	42	Figure 1, Figure 3, Table 3, Supplementary Table S1	100% Exact Match across all 5 purities (30/30 caller-purity runs)	HG008 somatic tumor-purity titration (20%→1% purity / 10%→0.5% VAF) at -f 0.002 across 6 callers

[a]Assay classes span targeted clinical panels and amplicons [11], whole-exome sequencing [10, 20], 51.8 GiB transcriptome RNA-seq [13], 10-platform Chromosome 20 sequencing chemistries [6, 13, 19], and full-genome germline/somatic WGS [13–19, 21–25]. [b]Every experiment completed with process exit code 0 across 60/60 benchmark archive directories (exact Clopper-Pearson 95% CI 94.04%–100.00%) and maps directly to the figures and tables listed in Column 6.

Supplementary Table S2. Baseline Equivalence Suites and Complete 87-Arm Full-Core Intermediate (.var) & VCF Parity Sweep

Evaluation Suite	Sub-Arm / Configuration & Format[a]	VarDictJava [1] Rows / Variants	VarDictRust Rows / Variants	FP vs. Java	FN vs. Java	Parity Verdict[a]
Amplicon Baseline	EGFR/MRE11A targeted amplicon calling (-a 10:0.95) (VCF)	2	2	0	0	100% Exact Match
Somatic Benchmark Slice	NIST GIAB HG008 [19] tumor-normal somatic slice (VCF)	18,426	18,426	0	0	100% Exact Match
Multi-Region Reference Offset	GIAB HG002 [13, 18] Chr20 multi-region coordinate offset verification (VCF)	2,572	2,572	0	0	100% Exact Match
Tile-Size Invariance[b]	SEQC2 HCC1395 [10] halo-tiling invariance (0.5, 1, 2, 10 Mbp tiles) (VCF)	5,090	5,090	0	0	100% Exact Match (all 4 tile sizes)[b]
Lossless BAM-vs-CRAM Twin	GIAB HG008 [19] lossless BAM-vs-CRAM [26] twin & 18 multimodal slices (VCF)	1,129	1,129	0	0	100% Exact Match (0 byte differences)
4-Assay Regression Suite	Targeted EGFR/MRE11A amplicon (-a) (VCF)	2	2	0	0	100% Exact Match
4-Assay Regression Suite	GIAB HG002 [13, 18] deep WGS slice (VCF)	227,562	227,562	0	0	100% Exact Match
4-Assay Regression Suite	GIAB HG008 [19] tumor-normal somatic slice (VCF)	18,426	18,426	0	0	100% Exact Match
4-Assay Regression Suite	SEQC2 HCC1395 [10] paired somatic exome slice (VCF)	623,444	623,444	0	0	100% Exact Match

Evaluation Suite	Sub-Arm / Configuration & Format[a]	VarDictJava [1] Rows / Variants	VarDictRust Rows / Variants	FP vs. Java	FN vs. Java	Parity Verdict[a]
Contiguous 249 Mbp -R chr1 [11][c]	Unchunked -R chr1:1-249250621 single window (--tile-size 250000000) (.var)[c]	4,705	4,705	0	0	100% Exact Match[c]
GIAB HG002 Chr20 Exome [11, 20]	GIAB HG002 Chr20 exome [20] (.var)	8,330	8,330	0	0	100% Exact Match
GIAB HG002 Chr20 Exome [11, 20]	GIAB HG002 Chr20 exome [20] (VCF)	8,233	8,233	0	0	100% Exact Match
Multimodal & CRAM Twin Suite [26]	17-modality extended CLI coverage suite (.var & VCF)	17	17	0	0	100% Exact Match
Multimodal & CRAM Twin Suite [26]	18-slice multimodal regression suite (VCF)	18	18	0	0	100% Exact Match
Multimodal & CRAM Twin Suite [26]	GIAB HG008 [19] CRAM-vs-BAM [26] twin slice (VCF)	1	1	0	0	100% Exact Match
Multimodal & CRAM Twin Suite [26]	Lossless CRAM-vs-BAM [26] bitwise twin verification (VCF)	40	40	0	0	100% Exact Match
51.8 GiB HG002 RNA-Seq [13]	51.8 GiB HG002 GM24385 [13] splice junction table (-i) (.splice.tsv)	12,668	12,668	0	0	100% Exact Match
51.8 GiB HG002 RNA-Seq [13]	51.8 GiB HG002 GM24385 [13] cDNA variant calls (-i) (.var)	1,230	1,230	0	0	100% Exact Match
10 Mbp Chr1 All-Positions (-f 0)[c]	Chromosome 1 10 Mbp at -f 0 (--tile-size 10000000) (.var)[c]	7,077	7,077	0	0	100% Exact Match[c]
10 Mbp Chr1 All-Positions (-f 0)[c]	Chromosome 1 10 Mbp at -f 0 (--tile-size 10000000) (VCF)[c]	7,032	7,032	0	0	100% Exact Match[c]
12-Sub-Arm CLI Mode Matrix [11]	Simple germline calling (-f 0.01) (.var)	228	228	0	0	100% Exact Match
12-Sub-Arm CLI Mode Matrix [11]	Simple zero-threshold MRD calling (-f 0) (.var)	248	248	0	0	100% Exact Match
12-Sub-Arm CLI Mode Matrix [11]	Simple germline calling with native Fisher exact test (--fisher) (.var)	650	650	0	0	100% Exact Match
12-Sub-Arm CLI Mode Matrix [11]	Chromosome 17 targeted clinical panel (.var)	83	83	0	0	100% Exact Match
12-Sub-Arm CLI Mode Matrix [11]	Structural variant deletion regions (.var)	3	3	0	0	100% Exact Match
12-Sub-Arm CLI Mode Matrix [11]	Low-coverage germline boundary calling (.var)	146	146	0	0	100% Exact Match
12-Sub-Arm CLI Mode Matrix [11]	Chromosome 17 amplicon awareness (-a) (.var)	145	145	0	0	100% Exact Match
12-Sub-Arm CLI Mode Matrix [11]	Amplicon calling with native Fisher exact test (-a --fisher) (.var)	145	145	0	0	100% Exact Match
12-Sub-Arm CLI Mode Matrix [11]	Paired tumor-normal somatic calling (-f 0.01) (.var)	664	664	0	0	100% Exact Match
12-Sub-Arm CLI Mode Matrix [11]	Paired somatic zero-threshold MRD calling (-f 0) (.var)	1,396	1,396	0	0	100% Exact Match
12-Sub-Arm CLI Mode Matrix [11]	Paired somatic calling with native Fisher exact test (--fisher) (.var)	527	527	0	0	100% Exact Match
12-Sub-Arm CLI Mode Matrix [11]	CRAM-vs-BAM [26] lossless twin intermediate table (-G) (.var)	228	228	0	0	100% Exact Match
AML Panel SRR15006540 [11]	Clinical AML panel SRR15006540 [11] (.var)	16,031	16,031	0	0	100% Exact Match
AML Panel SRR15006540 [11]	Clinical AML panel SRR15006540 [11] (VCF)	15,912	15,912	0	0	100% Exact Match
AML Panel SRR8657348 [11]	Clinical AML panel SRR8657348 [11] (.var)	54,448	54,448	0	0	100% Exact Match
AML Panel SRR8657348 [11]	Clinical AML panel SRR8657348 [11] (VCF)	53,224	53,224	0	0	100% Exact Match
AML Somatic Pair (SRR15006376 / SRR15006386) [11]	Clinical AML somatic pair SRR15006376 / SRR15006386 [11] (.var)	14,126	14,126	0	0	100% Exact Match
AML Somatic Pair (SRR15006376 / SRR15006386) [11]	Clinical AML somatic pair SRR15006376 / SRR15006386 [11] (VCF)	14,000	14,000	0	0	100% Exact Match
Ultra-Deep MRD Stress Suite	SRR15006375 (-f 0.0005 --fisher) (.var)	2,780	2,780	0	0	100% Exact Match
Ultra-Deep MRD Stress Suite	SRR15006375 (-f 0.0005 --fisher) (VCF)	2,645	2,645	0	0	100% Exact Match
Ultra-Deep MRD Stress Suite	SRR15006375 (-f 0 --fisher) (.var)	6,182	6,182	0	0	100% Exact Match
Ultra-Deep MRD Stress Suite	SRR15006375 (-f 0 --fisher) (VCF)	5,685	5,685	0	0	100% Exact Match
Ultra-Deep MRD Stress Suite	SRR15006540 (-f 0.0005 --fisher) (.var)	5,046	5,046	0	0	100% Exact Match
Ultra-Deep MRD Stress Suite	SRR15006540 (-f 0.0005 --fisher) (VCF)	4,711	4,711	0	0	100% Exact Match
Ultra-Deep MRD Stress Suite	SRR15006540 (-f 0 --fisher) (.var)	7,044	7,044	0	0	100% Exact Match
Ultra-Deep MRD Stress Suite	SRR15006540 (-f 0 --fisher) (VCF)	6,313	6,313	0	0	100% Exact Match
Ultra-Deep MRD Stress Suite	SRR15006386 (-f 0.0005 --fisher) (.var)	3,028	3,028	0	0	100% Exact Match
Ultra-Deep MRD Stress Suite	SRR15006386 (-f 0.0005 --fisher) (VCF)	2,888	2,888	0	0	100% Exact Match

Evaluation Suite	Sub-Arm / Configuration & Format[a]	VarDictJava [1] Rows / Variants	VarDictRust Rows / Variants	FP vs. Java	FN vs. Java	Parity Verdict[a]
Ultra-Deep MRD Stress Suite	SRR15006386 (-f 0 --fisher) (.var)	6,153	6,153	0	0	100% Exact Match
Ultra-Deep MRD Stress Suite	SRR15006386 (-f 0 --fisher) (VCF)	5,608	5,608	0	0	100% Exact Match
Ultra-Deep MRD Stress Suite	SRR15006376 (-f 0.0005 --fisher) (.var)	2,844	2,844	0	0	100% Exact Match
Ultra-Deep MRD Stress Suite	SRR15006376 (-f 0.0005 --fisher) (VCF)	2,732	2,732	0	0	100% Exact Match
Ultra-Deep MRD Stress Suite	SRR15006376 (-f 0 --fisher) (.var)	3,841	3,841	0	0	100% Exact Match
Ultra-Deep MRD Stress Suite	SRR15006376 (-f 0 --fisher) (VCF)	3,630	3,630	0	0	100% Exact Match
Ultra-Deep MRD Stress Suite	AML Somatic SRR15006376 / SRR15006386 (29kx depth) (-f 0.0005 --fisher) (.var)	4,045	4,045	0	0	100% Exact Match
Ultra-Deep MRD Stress Suite	AML Somatic SRR15006376 / SRR15006386 (29kx depth) (-f 0.0005 --fisher) (VCF)	3,855	3,855	0	0	100% Exact Match
Ultra-Deep MRD Stress Suite	AML Somatic SRR15006376 / SRR15006386 (29kx depth) (-f 0 --fisher) (.var)	6,890	6,890	0	0	100% Exact Match
Ultra-Deep MRD Stress Suite	AML Somatic SRR15006376 / SRR15006386 (29kx depth) (-f 0 --fisher) (VCF)	6,309	6,309	0	0	100% Exact Match
Ultra-Deep MRD Stress Suite	AML Somatic SRR15006375 / SRR15006386 (45kx depth) (-f 0.0005 --fisher) (.var)	4,481	4,481	0	0	100% Exact Match
Ultra-Deep MRD Stress Suite	AML Somatic SRR15006375 / SRR15006386 (45kx depth) (-f 0.0005 --fisher) (VCF)	4,262	4,262	0	0	100% Exact Match
Ultra-Deep MRD Stress Suite	AML Somatic SRR15006375 / SRR15006386 (45kx depth) (-f 0 --fisher) (.var)	7,474	7,474	0	0	100% Exact Match
Ultra-Deep MRD Stress Suite	AML Somatic SRR15006375 / SRR15006386 (45kx depth) (-f 0 --fisher) (VCF)	6,828	6,828	0	0	100% Exact Match
Ultra-Deep MRD Stress Suite	GIAB HG008 [19] Somatic (15kx depth) (-f 0.0005 --fisher) (.var)	19	19	0	0	100% Exact Match
Ultra-Deep MRD Stress Suite	GIAB HG008 [19] Somatic (15kx depth) (-f 0.0005 --fisher) (VCF)	16	16	0	0	100% Exact Match
Ultra-Deep MRD Stress Suite	GIAB HG002 [13] Germline (7kx depth) (-f 0.0005 --fisher) (.var)	1,008	1,008	0	0	100% Exact Match
Ultra-Deep MRD Stress Suite	GIAB HG002 [13] Germline (7kx depth) (-f 0.0005 --fisher) (VCF)	640	640	0	0	100% Exact Match
Ultra-Deep MRD Stress Suite	HG002 GM24385 [13] RNA-Seq (173kx depth) (-f 0.001 --fisher) (.var)	139	139	0	0	100% Exact Match
Ultra-Deep MRD Stress Suite	HG002 GM24385 [13] RNA-Seq (173kx depth) (-f 0.001 --fisher) (VCF)	120	120	0	0	100% Exact Match
WGS & WES Low-VAF MRD Matrix	GIAB HG002 [13, 18] WGS MRD (-f 0.001) (.var)	15,083	15,083	0	0	100% Exact Match
WGS & WES Low-VAF MRD Matrix	GIAB HG002 [13, 18] WGS MRD (-f 0.001) (VCF)	14,769	14,769	0	0	100% Exact Match
WGS & WES Low-VAF MRD Matrix	GIAB HG002 [13, 18] WGS MRD (-f 0) (.var)	15,083	15,083	0	0	100% Exact Match
WGS & WES Low-VAF MRD Matrix	GIAB HG002 [13, 18] WGS MRD (-f 0) (VCF)	14,769	14,769	0	0	100% Exact Match
WGS & WES Low-VAF MRD Matrix	GIAB HG008 [19] Somatic WGS MRD (-f 0.001) (.var)	21,379	21,379	0	0	100% Exact Match
WGS & WES Low-VAF MRD Matrix	GIAB HG008 [19] Somatic WGS MRD (-f 0.001) (VCF)	20,293	20,293	0	0	100% Exact Match
WGS & WES Low-VAF MRD Matrix	GIAB HG008 [19] Somatic WGS MRD (-f 0.001 --fisher) (.var)	21,379	21,379	0	0	100% Exact Match
WGS & WES Low-VAF MRD Matrix	GIAB HG008 [19] Somatic WGS MRD (-f 0.001 --fisher) (VCF)	20,293	20,293	0	0	100% Exact Match
WGS & WES Low-VAF MRD Matrix	GIAB HG008 [19] Somatic WGS MRD (-f 0) (.var)	21,379	21,379	0	0	100% Exact Match
WGS & WES Low-VAF MRD Matrix	GIAB HG008 [19] Somatic WGS MRD (-f 0) (VCF)	20,293	20,293	0	0	100% Exact Match
WGS & WES Low-VAF MRD Matrix	GIAB HG005 [13] 300x WGS chr3 (-f 0.001) (.var)	339	339	0	0	100% Exact Match
WGS & WES Low-VAF MRD Matrix	GIAB HG005 [13] 300x WGS chr3 (-f 0.001) (VCF)	325	325	0	0	100% Exact Match
WGS & WES Low-VAF MRD Matrix	GIAB HG005 [13] 300x WGS chr3 (-f 0) (.var)	339	339	0	0	100% Exact Match
WGS & WES Low-VAF MRD Matrix	GIAB HG005 [13] 300x WGS chr3 (-f 0) (VCF)	325	325	0	0	100% Exact Match

Evaluation Suite	Sub-Arm / Configuration & Format[a]	VarDictJava [1] Rows / Variants	VarDictRust Rows / Variants	FP vs. Java	FN vs. Java	Parity Verdict[a]
WGS & WES Low-VAF MRD Matrix	GIAB HG005 [13] 300x WGS chr6 (-f 0.001) (.var)	305	305	0	0	100% Exact Match
WGS & WES Low-VAF MRD Matrix	GIAB HG005 [13] 300x WGS chr6 (-f 0.001) (VCF)	299	299	0	0	100% Exact Match
WGS & WES Low-VAF MRD Matrix	GIAB HG005 [13] 300x WGS chr21 (-f 0.001) (.var)	286	286	0	0	100% Exact Match
WGS & WES Low-VAF MRD Matrix	GIAB HG005 [13] 300x WGS chr21 (-f 0.001) (VCF)	284	284	0	0	100% Exact Match
WGS & WES Low-VAF MRD Matrix	SEQC2 HCC1395 [10] Somatic Exome MRD (-f 0.001) (.var)	5,261	5,261	0	0	100% Exact Match
WGS & WES Low-VAF MRD Matrix	SEQC2 HCC1395 [10] Somatic Exome MRD (-f 0.001) (VCF)	5,206	5,206	0	0	100% Exact Match
WGS & WES Low-VAF MRD Matrix	SEQC2 HCC1395 [10] Somatic Exome MRD (-f 0.001 --fisher) (.var)	5,261	5,261	0	0	100% Exact Match
WGS & WES Low-VAF MRD Matrix	SEQC2 HCC1395 [10] Somatic Exome MRD (-f 0.001 --fisher) (VCF)	5,206	5,206	0	0	100% Exact Match
WGS & WES Low-VAF MRD Matrix	SEQC2 HCC1395 [10] Somatic Exome MRD (-f 0) (.var)	5,261	5,261	0	0	100% Exact Match
WGS & WES Low-VAF MRD Matrix	SEQC2 HCC1395 [10] Somatic Exome MRD (-f 0) (VCF)	5,206	5,206	0	0	100% Exact Match

[a]All 87/87 (100.00%) comparative sub-arms across all 12 multi-regime sweep suites and all 5 baseline equivalence suites achieve 100% exact intermediate (.var, .splice.tsv) and VCF parity against VarDictJava v1.8.3 [1] (0 false positives and 0 false negatives at eps=1e-6). [b]Halo-tiling invariance confirms identical variant output across 0.5, 1.0, 2.0, and 10.0 Mbp tile sizes on SEQC2 HCC1395 [10]. [c]Standard BED intervals (<=1 Mbp) run untiled by default; the two multi-megabase single-interval stress suites (10 Mbp chr1 at -f 0 and 249.25 Mbp -R chr1) use --tile-size matching the region span so that single-window VarDictRust is compared directly against single-window VarDictJava [1].

Supplementary Table S3. Complete Technical Replicate (n=3) and Full-WGS Wall-Clock and Peak RSS Ledger

Benchmark Suite	Workload / Cohort	Threads	Caller / Engine	Replicate Wall Times (s)[a]	Wall Mean (Min–Max) (s)[a]	Replicate Peak RSS (KB)[a]	Peak RSS Mean (MB / GiB)
AML Panel Replication [11]	AML Panel SRR15006375	1	VarDictRust	121.14, 122.26, 122.38	121.93 (121.14–122.38)	215,108, 215,236, 215,736	210.3 MB (0.205 GiB)
AML Panel Replication [11]	AML Panel SRR15006375	1	VarDictJava [1]	404.89, 418.94, 400.74	408.19 (400.74–418.94)	2,590,208, 2,591,732, 2,612,224	2537.2 MB (2.478 GiB)
AML Panel Replication [11]	AML Panel SRR15006375	1	vardictcpp [11]	109.90, 109.88, 109.19	109.66 (109.19–109.90)	65,292, 65,280, 65,312	63.8 MB (0.062 GiB)
AML Panel Replication [11]	AML Panel SRR15006375	8	VarDictRust	49.39, 49.59, 49.44	49.47 (49.39–49.59)	278,052, 274,168, 277,424	270.1 MB (0.264 GiB)
AML Panel Replication [11]	AML Panel SRR15006375	8	VarDictJava [1]	229.14, 235.27, 233.69	232.70 (229.14–235.27)	4,665,344, 4,511,724, 4,371,936	4410.5 MB (4.307 GiB)
AML Panel Replication [11]	AML Panel SRR15006375	8	vardictcpp [11]	15.42, 15.32, 15.30	15.35 (15.30–15.42)	342,664, 334,764, 331,092	328.3 MB (0.321 GiB)
AML Panel Replication [11]	AML Panel SRR15006376	1	VarDictRust	80.84, 80.32, 80.08	80.41 (80.08–80.84)	231,180, 231,156, 230,976	225.7 MB (0.220 GiB)
AML Panel Replication [11]	AML Panel SRR15006376	1	VarDictJava [1]	326.10, 340.26, 336.12	334.16 (326.10–340.26)	4,428,032, 4,031,016, 4,579,804	4244.4 MB (4.145 GiB)
AML Panel Replication [11]	AML Panel SRR15006376	1	vardictcpp [11]	83.78, 84.32, 83.03	83.71 (83.03–84.32)	75,804, 75,808, 75,804	74.0 MB (0.072 GiB)
AML Panel Replication [11]	AML Panel SRR15006376	8	VarDictRust	48.86, 48.70, 48.65	48.74 (48.65–48.86)	287,200, 288,108, 287,872	281.0 MB (0.274 GiB)
AML Panel Replication [11]	AML Panel SRR15006376	8	VarDictJava [1]	174.29, 150.76, 168.04	164.36 (150.76–174.29)	7,744,512, 8,044,032, 7,073,792	7442.2 MB (7.268 GiB)
AML Panel Replication [11]	AML Panel SRR15006376	8	vardictcpp [11]	13.09, 12.50, 12.96	12.85 (12.50–13.09)	351,772, 364,060, 371,688	354.0 MB (0.346 GiB)
AML Panel Replication [11]	AML Panel SRR15006386	1	VarDictRust	114.04, 113.35, 113.17	113.52 (113.17–114.04)	191,748, 191,976, 192,300	187.5 MB (0.183 GiB)
AML Panel Replication [11]	AML Panel SRR15006386	1	VarDictJava [1]	334.16, 338.80, 331.87	334.94 (331.87–338.80)	2,596,316, 2,423,020, 2,425,344	2423.4 MB (2.367 GiB)
AML Panel Replication [11]	AML Panel SRR15006386	1	vardictcpp [11]	98.37, 98.14, 98.09	98.20 (98.09–98.37)	57,516, 57,516, 57,520	56.2 MB (0.055 GiB)
AML Panel Replication [11]	AML Panel SRR15006386	8	VarDictRust	43.81, 43.88, 43.64	43.78 (43.64–43.88)	312,272, 312,860, 313,136	305.4 MB (0.298 GiB)
AML Panel Replication [11]	AML Panel SRR15006386	8	VarDictJava [1]	178.99, 182.17, 180.70	180.62 (178.99–182.17)	4,535,524, 4,536,296, 4,551,680	4434.7 MB (4.331 GiB)

Benchmark Suite	Workload / Cohort	Threads	Caller / Engine	Replicate Wall Times (s)[a]	Wall Mean (Min–Max) (s)[a]	Replicate Peak RSS (KB)[a]	Peak RSS Mean (MB / GiB)
AML Panel Replication [11]	AML Panel SRR15006386	8	vardictcpp [11]	16.69, 15.01, 14.95	15.55 (14.95–16.69)	300,084, 309,872, 307,332	298.6 MB (0.292 GiB)
AML Panel Replication [11]	AML Panel SRR15006540	1	VarDictRust	125.49, 125.61, 126.91	126.00 (125.49–126.91)	199,412, 199,396, 199,072	194.6 MB (0.190 GiB)
AML Panel Replication [11]	AML Panel SRR15006540	1	VarDictJava [1]	406.72, 412.42, 402.18	407.11 (402.18–412.42)	4,630,248, 4,671,184, 4,745,216	4572.5 MB (4.465 GiB)
AML Panel Replication [11]	AML Panel SRR15006540	1	vardictcpp [11]	111.55, 111.32, 111.66	111.51 (111.32–111.66)	78,084, 78,088, 78,084	76.3 MB (0.074 GiB)
AML Panel Replication [11]	AML Panel SRR15006540	8	VarDictRust	77.89, 77.99, 78.24	78.04 (77.89–78.24)	233,572, 227,592, 229,364	224.8 MB (0.220 GiB)
AML Panel Replication [11]	AML Panel SRR15006540	8	VarDictJava [1]	179.54, 186.40, 187.21	184.38 (179.54–187.21)	3,863,552, 4,506,172, 4,187,860	4087.8 MB (3.992 GiB)
AML Panel Replication [11]	AML Panel SRR15006540	8	vardictcpp [11]	16.77, 16.31, 17.00	16.69 (16.31–17.00)	342,792, 350,980, 347,656	339.0 MB (0.331 GiB)
AML Panel Replication [11]	AML Panel SRR8657348	1	VarDictRust	155.05, 155.12, 154.77	154.98 (154.77–155.12)	486,840, 486,476, 483,624	474.3 MB (0.463 GiB)
AML Panel Replication [11]	AML Panel SRR8657348	1	VarDictJava [1]	862.51, 857.96, 863.07	861.18 (857.96–863.07)	2,079,488, 2,078,976, 2,073,724	2028.7 MB (1.981 GiB)
AML Panel Replication [11]	AML Panel SRR8657348	1	vardictcpp [11]	175.43, 173.98, 174.02	174.48 (173.98–175.43)	146,848, 146,932, 146,936	143.5 MB (0.140 GiB)
AML Panel Replication [11]	AML Panel SRR8657348	8	VarDictRust	39.76, 39.62, 39.50	39.63 (39.50–39.76)	492,940, 485,740, 536,324	493.2 MB (0.482 GiB)
AML Panel Replication [11]	AML Panel SRR8657348	8	VarDictJava [1]	212.70, 217.72, 217.12	215.85 (212.70–217.72)	3,713,444, 3,823,360, 3,822,592	3697.7 MB (3.611 GiB)
AML Panel Replication [11]	AML Panel SRR8657348	8	vardictcpp [11]	24.71, 24.68, 24.64	24.68 (24.64–24.71)	255,472, 259,568, 258,544	251.8 MB (0.246 GiB)
AML Panel Replication [11]	AML Panel Somatic Pair (SRR15006376 / SRR15006386)	1	VarDictRust	230.96, 230.70, 229.92	230.53 (229.92–230.96)	255,464, 252,304, 253,316	247.8 MB (0.242 GiB)
AML Panel Replication [11]	AML Panel Somatic Pair (SRR15006376 / SRR15006386)	1	VarDictJava [1]	720.47, 728.53, 729.27	726.09 (720.47–729.27)	4,415,396, 4,396,032, 4,420,820	4307.4 MB (4.206 GiB)
AML Panel Replication [11]	AML Panel Somatic Pair (SRR15006376 / SRR15006386)	1	vardictcpp [11]	215.50, 215.04, 214.94	215.16 (214.94–215.50)	148,252, 148,252, 148,504	144.9 MB (0.141 GiB)
AML Panel Replication [11]	AML Panel Somatic Pair (SRR15006376 / SRR15006386)	8	VarDictRust	134.96, 133.83, 137.51	135.43 (133.83–137.51)	383,724, 396,596, 393,488	382.1 MB (0.373 GiB)
AML Panel Replication [11]	AML Panel Somatic Pair (SRR15006376 / SRR15006386)	8	VarDictJava [1]	373.79, 371.55, 374.68	373.34 (371.55–374.68)	5,850,600, 5,814,784, 5,783,296	5679.9 MB (5.547 GiB)
AML Panel Replication [11]	AML Panel Somatic Pair (SRR15006376 / SRR15006386)	8	vardictcpp [11]	31.60, 31.64, 31.75	31.66 (31.60–31.75)	824,788, 803,560, 786,332	786.0 MB (0.768 GiB)
Unchunked Chromosome 1 (-R chr1) [11]	SRR8657348 Unchunked -R chr1 (249.25 Mbp)	8	VarDictJava [1] (-Xmx200g)	509.65, 492.93, 502.90	501.83 (492.93–509.65)	112,714,496, 111,591,168, 110,633,216	109029.6 MB (106.474 GiB)
Unchunked Chromosome 1 (-R chr1) [11]	SRR8657348 Unchunked -R chr1 (249.25 Mbp)	8	VarDictRust	9.80, 9.81, 9.64	9.75 (9.64–9.81)	320,192, 319,576, 339,012	318.6 MB (0.311 GiB)
Unchunked Chromosome 1 (-R chr1) [11]	SRR8657348 Unchunked -R chr1 (249.25 Mbp)	8	vardictcpp [11]	106.58, 105.07, 106.03	105.89 (105.07–106.58)	11,098,040, 11,098,040, 11,098,040	10837.9 MB (10.584 GiB)
Somatic Field Callers [11][b]	AML Somatic Pair (SRR15006376 / SRR15006386)	8	VarDictRust	135.43	135.43 (single run)[b]	391,269	382.1 MB (0.373 GiB)
Somatic Field Callers [11][b]	AML Somatic Pair (SRR15006376 / SRR15006386)	8	VarDictJava [1]	373.34	373.34 (single run)[b]	5,816,227	5679.9 MB (5.547 GiB)
Somatic Field Callers [11][b]	AML Somatic Pair (SRR15006376 / SRR15006386)	8	vardictcpp [11]	31.66	31.66 (single run)[b]	804,893	786.0 MB (0.768 GiB)
Somatic Field Callers [11][b]	AML Somatic Pair (SRR15006376 / SRR15006386)	8	Strelka2 [4]	199.96	199.96 (single run)[b]	1,184,452	1156.7 MB (1.130 GiB)
Somatic Field Callers [11][b]	AML Somatic Pair (SRR15006376 / SRR15006386)	8	LoFreq [7]	521.82	521.82 (single run)[b]	2,146,108	2095.8 MB (2.047 GiB)
Somatic Field Callers [11][b]	AML Somatic Pair (SRR15006376 / SRR15006386)	8	Octopus [5]	450.87	450.87 (single run)[b]	2,521,004	2461.9 MB (2.404 GiB)
Somatic Field Callers [11][b]	AML Somatic Pair (SRR15006376 / SRR15006386)	8	Mutect2 [8]	969.01	969.01 (single run)[b]	2,355,632	2300.4 MB (2.247 GiB)
Transcriptome RNA-Seq Splice (-i)	51.8 GiB GIAB HG002 GM24385 [13] RNA-Seq (-i)	8	VarDictRust	5.01, 4.90, 4.90	4.94 (4.90–5.01)	361,360, 368,040, 376,072	359.9 MB (0.351 GiB)
Transcriptome RNA-Seq Splice (-i)	51.8 GiB GIAB HG002 GM24385 [13] RNA-Seq (-i)	8	VarDictJava [1]	23.21, 23.03, 23.70	23.31 (23.03–23.70)	5,324,544, 4,538,880, 5,217,280	4909.1 MB (4.794 GiB)

Benchmark Suite	Workload / Cohort	Threads	Caller / Engine	Replicate Wall Times (s)[a]	Wall Mean (Min–Max) (s)[a]	Replicate Peak RSS (KB)[a]	Peak RSS Mean (MB / GiB)
4-Target Performance Budget	Targeted EGFR/MRE11A Amplicon Panel	8	VarDictRust	0.01, 0.01, 0.01	0.01 (0.01–0.01)	11,776, 11,776, 12,032	11.6 MB (0.011 GiB)
4-Target Performance Budget	Targeted EGFR/MRE11A Amplicon Panel	8	VarDictJava [1]	3.06, 3.12, 3.07	3.08 (3.06–3.12)	122,744, 122,968, 122,964	120.0 MB (0.117 GiB)
4-Target Performance Budget	Targeted EGFR/MRE11A Amplicon Panel	8	vardictcpp [11]	5.37, 5.22, 5.24	5.28 (5.22–5.37)	122,600, 122,576, 122,572	119.7 MB (0.117 GiB)
4-Target Performance Budget	GIAB HG002 [13, 18] Germline Benchmark Slice	8	VarDictRust	21.30, 21.33, 21.29	21.31 (21.29–21.33)	486,236, 519,872, 487,080	486.1 MB (0.475 GiB)
4-Target Performance Budget	GIAB HG002 [13, 18] Germline Benchmark Slice	8	VarDictJava [1]	159.59, 152.13, 153.25	154.99 (152.13–159.59)	10,989,804, 11,341,096, 10,761,680	10772.3 MB (10.520 GiB)
4-Target Performance Budget	GIAB HG002 [13, 18] Germline Benchmark Slice	8	vardictcpp [11]	21.14, 20.69, 20.10	20.64 (20.10–21.14)	879,700, 880,956, 881,460	860.1 MB (0.840 GiB)
4-Target Performance Budget	GIAB HG008 [19] Somatic Benchmark Slice	8	VarDictRust	17.14, 17.19, 17.11	17.15 (17.11–17.19)	522,756, 518,000, 528,268	510.8 MB (0.499 GiB)
4-Target Performance Budget	GIAB HG008 [19] Somatic Benchmark Slice	8	VarDictJava [1]	79.15, 78.54, 77.28	78.32 (77.28–79.15)	17,866,968, 18,674,616, 17,734,080	17667.9 MB (17.254 GiB)
4-Target Performance Budget	GIAB HG008 [19] Somatic Benchmark Slice	8	vardictcpp [11]	17.30, 17.17, 16.91	17.13 (16.91–17.30)	578,696, 584,836, 582,644	568.4 MB (0.555 GiB)
4-Target Performance Budget	SEQC2 HCC1395 [10] Paired Somatic Exome Slice	8	VarDictRust	279.29, 277.68, 277.52	278.16 (277.52–279.29)	526,640, 526,944, 529,060	515.2 MB (0.503 GiB)
4-Target Performance Budget	SEQC2 HCC1395 [10] Paired Somatic Exome Slice	8	VarDictJava [1]	3967.00, 4030.00, 4045.00	4014.00 (3967.00–4045.00)	14,618,112, 14,880,508, 16,471,012	14964.1 MB (14.613 GiB)
4-Target Performance Budget	SEQC2 HCC1395 [10] Paired Somatic Exome Slice	8	vardictcpp [11]	293.71, 294.67, 287.70	292.03 (287.70–294.67)	2,314,000, 2,274,252, 2,299,560	2242.1 MB (2.190 GiB)
Whole-Exome Thread Scaling	SEQC2 HCC1395 [10] Single-Sample Whole Exome	1	VarDictRust	805.74, 806.69, 825.81	812.75 (805.74–825.81)	351,724, 351,844, 351,652	343.5 MB (0.335 GiB)
Whole-Exome Thread Scaling	SEQC2 HCC1395 [10] Single-Sample Whole Exome	1	VarDictJava [1]	5767.00, 5789.00, 5717.00	5757.67 (5717.00–5789.00)	1,857,212, 1,887,008, 1,903,816	1838.5 MB (1.795 GiB)
Whole-Exome Thread Scaling	SEQC2 HCC1395 [10] Single-Sample Whole Exome	1	vardictcpp [11]	993.97, 1010.80, 1012.37	1005.71 (993.97–1012.37)	1,264,172, 1,263,912, 1,264,168	1234.5 MB (1.206 GiB)
Whole-Exome Thread Scaling	SEQC2 HCC1395 [10] Single-Sample Whole Exome	2	VarDictRust	405.60, 411.13, 410.70	409.14 (405.60–411.13)	428,676, 428,772, 417,680	415.1 MB (0.405 GiB)
Whole-Exome Thread Scaling	SEQC2 HCC1395 [10] Single-Sample Whole Exome	2	VarDictJava [1]	5762.00, 5839.00, 5806.00	5802.33 (5762.00–5839.00)	5,351,168, 4,222,464, 5,321,168	4848.6 MB (4.735 GiB)
Whole-Exome Thread Scaling	SEQC2 HCC1395 [10] Single-Sample Whole Exome	2	vardictcpp [11]	529.51, 530.65, 531.93	530.70 (529.51–531.93)	1,357,128, 1,357,144, 1,357,020	1325.3 MB (1.294 GiB)
Whole-Exome Thread Scaling	SEQC2 HCC1395 [10] Single-Sample Whole Exome	4	VarDictRust	211.15, 212.32, 213.44	212.30 (211.15–213.44)	462,724, 454,340, 444,064	443.1 MB (0.433 GiB)
Whole-Exome Thread Scaling	SEQC2 HCC1395 [10] Single-Sample Whole Exome	4	VarDictJava [1]	2254.41, 2276.68, 2280.43	2270.51 (2254.41–2280.43)	6,932,480, 7,551,728, 6,877,428	6953.7 MB (6.791 GiB)
Whole-Exome Thread Scaling	SEQC2 HCC1395 [10] Single-Sample Whole Exome	4	vardictcpp [11]	267.51, 267.11, 267.34	267.32 (267.11–267.51)	1,438,412, 1,434,796, 1,437,096	1403.1 MB (1.370 GiB)
Whole-Exome Thread Scaling	SEQC2 HCC1395 [10] Single-Sample Whole Exome	8	VarDictRust	117.36, 119.47, 119.88	118.90 (117.36–119.88)	546,844, 553,576, 548,892	536.9 MB (0.524 GiB)
Whole-Exome Thread Scaling	SEQC2 HCC1395 [10] Single-Sample Whole Exome	8	VarDictJava [1]	1752.96, 1911.57, 1779.37	1814.63 (1752.96–1911.57)	9,542,144, 9,317,612, 9,601,380	9264.7 MB (9.048 GiB)
Whole-Exome Thread Scaling	SEQC2 HCC1395 [10] Single-Sample Whole Exome	8	vardictcpp [11]	138.32, 138.71, 140.40	139.14 (138.32–140.40)	1,571,092, 1,571,564, 1,557,576	1530.0 MB (1.494 GiB)
Whole-Exome Thread Scaling	SEQC2 HCC1395 [10] Single-Sample Whole Exome	16	VarDictRust	75.89, 76.76, 74.66	75.77 (74.66–76.76)	613,520, 618,784, 633,916	607.5 MB (0.593 GiB)
Whole-Exome Thread Scaling	SEQC2 HCC1395 [10] Single-Sample Whole Exome	16	VarDictJava [1]	1885.00, 1929.42, 1866.57	1893.66 (1866.57–1929.42)	18,610,904, 17,986,784, 18,267,580	17859.8 MB (17.441 GiB)
Whole-Exome Thread Scaling	SEQC2 HCC1395 [10] Single-Sample Whole Exome	16	vardictcpp [11]	73.64, 73.74, 75.12	74.17 (73.64–75.12)	1,760,284, 1,774,788, 1,777,884	1729.5 MB (1.689 GiB)
Whole-Exome Thread Scaling	SEQC2 HCC1395 [10] Single-Sample Whole Exome	32	VarDictRust	67.27, 68.20, 68.10	67.86 (67.27–68.20)	763,156, 761,796, 770,404	747.2 MB (0.730 GiB)
Whole-Exome Thread Scaling	SEQC2 HCC1395 [10] Single-Sample Whole Exome	32	VarDictJava [1]	1969.99, 2001.39, 2003.54	1991.64 (1969.99–2003.54)	23,221,636, 23,172,288, 22,081,264	22290.1 MB (21.768 GiB)

Benchmark Suite	Workload / Cohort	Threads	Caller / Engine	Replicate Wall Times (s)[a]	Wall Mean (Min–Max) (s)[a]	Replicate Peak RSS (KB)[a]	Peak RSS Mean (MB / GiB)
Whole-Exome Thread Scaling	SEQC2 HCC1395 [10] Single-Sample Whole Exome	32	vardictcpp [11]	63.41, 64.15, 63.49	63.68 (63.41–64.15)	2,151,852, 2,164,356, 2,164,628	2109.7 MB (2.060 GiB)
Chromosome 1 Contiguous Span Scaling	Chromosome 1 Contiguous Span (1 Mbp)	8	VarDictRust	1.16, 1.17, 1.20	1.18 (1.16–1.20)	532,652, 531,732, 531,972	519.6 MB (0.507 GiB)
Chromosome 1 Contiguous Span Scaling	Chromosome 1 Contiguous Span (1 Mbp)	8	vardictcpp [11]	5.50, 5.45, 5.48	5.48 (5.45–5.50)	195,336, 195,592, 195,080	190.8 MB (0.186 GiB)
Chromosome 1 Contiguous Span Scaling	Chromosome 1 Contiguous Span (10 Mbp)	8	VarDictRust	1.60, 1.60, 1.60	1.60 (1.60–1.60)	1,243,864, 1,184,828, 1,264,216	1202.1 MB (1.174 GiB)
Chromosome 1 Contiguous Span Scaling	Chromosome 1 Contiguous Span (10 Mbp)	8	vardictcpp [11]	6.73, 6.67, 6.66	6.69 (6.66–6.73)	384,668, 386,380, 384,580	376.2 MB (0.367 GiB)
Chromosome 1 Contiguous Span Scaling	Chromosome 1 Contiguous Span (100 Mbp)	8	VarDictRust	12.20, 11.98, 11.80	11.99 (11.80–12.20)	1,277,644, 1,245,396, 1,311,332	1248.2 MB (1.219 GiB)
Chromosome 1 Contiguous Span Scaling	Chromosome 1 Contiguous Span (100 Mbp)	8	vardictcpp [11]	16.45, 16.32, 16.36	16.38 (16.32–16.45)	489,044, 456,176, 494,436	468.6 MB (0.458 GiB)
Chromosome 1 Contiguous Span Scaling	Chromosome 1 Contiguous Span (249.25 Mbp)	8	VarDictRust	27.18, 27.21, 27.77	27.39 (27.18–27.77)	1,455,976, 1,458,324, 1,428,280	1413.6 MB (1.380 GiB)
Chromosome 1 Contiguous Span Scaling	Chromosome 1 Contiguous Span (249.25 Mbp)	8	vardictcpp [11]	30.39, 30.33, 30.27	30.33 (30.27–30.39)	857,536, 814,916, 821,808	811.9 MB (0.793 GiB)
Chromosome 1 Contiguous Span Scaling	Chromosome 1 Contiguous Span (1 Mbp)	16	VarDictRust	1.14, 1.12, 1.12	1.13 (1.12–1.14)	532,756, 533,136, 532,956	520.5 MB (0.508 GiB)
Chromosome 1 Contiguous Span Scaling	Chromosome 1 Contiguous Span (1 Mbp)	16	vardictcpp [11]	5.13, 5.09, 5.16	5.13 (5.09–5.16)	354,976, 352,696, 354,120	345.6 MB (0.338 GiB)
Chromosome 1 Contiguous Span Scaling	Chromosome 1 Contiguous Span (10 Mbp)	16	VarDictRust	0.94, 0.94, 0.96	0.95 (0.94–0.96)	1,799,560, 1,909,712, 1,909,552	1829.0 MB (1.786 GiB)
Chromosome 1 Contiguous Span Scaling	Chromosome 1 Contiguous Span (10 Mbp)	16	vardictcpp [11]	5.50, 5.76, 5.83	5.70 (5.50–5.83)	690,112, 690,496, 694,340	675.4 MB (0.660 GiB)
Chromosome 1 Contiguous Span Scaling	Chromosome 1 Contiguous Span (100 Mbp)	16	VarDictRust	6.23, 6.23, 6.38	6.28 (6.23–6.38)	2,036,552, 2,025,764, 2,006,076	1975.4 MB (1.929 GiB)
Chromosome 1 Contiguous Span Scaling	Chromosome 1 Contiguous Span (100 Mbp)	16	vardictcpp [11]	10.75, 10.72, 10.56	10.68 (10.56–10.75)	836,088, 828,196, 814,352	806.9 MB (0.788 GiB)
Chromosome 1 Contiguous Span Scaling	Chromosome 1 Contiguous Span (249.25 Mbp)	16	VarDictRust	14.60, 15.14, 14.81	14.85 (14.60–15.14)	2,439,756, 2,382,796, 2,433,268	2361.9 MB (2.307 GiB)
Chromosome 1 Contiguous Span Scaling	Chromosome 1 Contiguous Span (249.25 Mbp)	16	vardictcpp [11]	17.70, 18.57, 18.90	18.39 (17.70–18.90)	1,209,508, 1,203,464, 1,169,436	1166.2 MB (1.139 GiB)
Chromosome 1 Contiguous Span Scaling	Chromosome 1 Contiguous Span (1 Mbp)	32	VarDictRust	1.17, 1.17, 1.16	1.17 (1.16–1.17)	535,504, 533,984, 535,260	522.4 MB (0.510 GiB)
Chromosome 1 Contiguous Span Scaling	Chromosome 1 Contiguous Span (1 Mbp)	32	vardictcpp [11]	5.39, 5.42, 5.55	5.45 (5.39–5.55)	399,284, 397,120, 400,828	389.7 MB (0.381 GiB)
Chromosome 1 Contiguous Span Scaling	Chromosome 1 Contiguous Span (10 Mbp)	32	VarDictRust	0.91, 0.92, 0.91	0.91 (0.91–0.92)	2,788,972, 2,733,968, 2,726,516	2685.4 MB (2.622 GiB)
Chromosome 1 Contiguous Span Scaling	Chromosome 1 Contiguous Span (10 Mbp)	32	vardictcpp [11]	6.04, 6.10, 6.16	6.10 (6.04–6.16)	1,262,680, 1,249,556, 1,240,044	1221.5 MB (1.193 GiB)
Chromosome 1 Contiguous Span Scaling	Chromosome 1 Contiguous Span (100 Mbp)	32	VarDictRust	5.34, 5.38, 5.16	5.29 (5.16–5.38)	2,946,828, 3,016,304, 3,043,928	2932.0 MB (2.863 GiB)
Chromosome 1 Contiguous Span Scaling	Chromosome 1 Contiguous Span (100 Mbp)	32	vardictcpp [11]	10.79, 10.30, 10.00	10.36 (10.00–10.79)	1,443,032, 1,493,356, 1,510,604	1447.6 MB (1.414 GiB)
Chromosome 1 Contiguous Span Scaling	Chromosome 1 Contiguous Span (249.25 Mbp)	32	VarDictRust	11.90, 11.96, 11.76	11.87 (11.76–11.96)	3,828,452, 3,939,028, 3,823,184	3773.0 MB (3.685 GiB)
Chromosome 1 Contiguous Span Scaling	Chromosome 1 Contiguous Span (249.25 Mbp)	32	vardictcpp [11]	16.17, 16.25, 16.26	16.23 (16.17–16.26)	1,821,828, 1,762,144, 1,756,708	1738.5 MB (1.698 GiB)
All-Positions Pileup (-f 0)	Chromosome 1 10 Mbp All-Positions (-f 0)	8	VarDictRust	20.09, 19.41, 19.24	19.58 (19.24–20.09)	6,185,864, 6,185,476, 6,185,552	6040.6 MB (5.899 GiB)
All-Positions Pileup (-f 0)	Chromosome 1 10 Mbp All-Positions (-f 0)	8	vardictcpp [11]	6.04, 5.99, 5.99	6.01 (5.99–6.04)	386,612, 380,596, 386,420	375.5 MB (0.367 GiB)
Concurrent 2x4-Thread Node Packing	SEQC2 [10] Whole Exome (Concurrent Job 1)	4	VarDictRust	212.56, 209.64, 211.94	211.38 (209.64–212.56)	449,244, 446,504, 448,288	437.5 MB (0.427 GiB)
Concurrent 2x4-Thread Node Packing	SEQC2 [10] Whole Exome (Concurrent Job 2)	4	VarDictRust	212.43, 209.86, 210.85	211.05 (209.86–212.43)	453,764, 451,428, 452,540	442.0 MB (0.432 GiB)
Full-Genome WGS Scaling (chr1–chr22)[c]	GIAB HG002 [13, 18] Full WGS (22 Unchunked Chromosomes)	32	VarDictRust	1516.61	25:16.61 (1,516.61 s)[c]	4,928,384	4.93 GB (4.70 GiB)
Full-Genome WGS Scaling (chr1–chr22)[c]	GIAB HG002 [13, 18] Full WGS (≤50 kbp GIAB Callable BED)	32	VarDictRust	1221.21	20:21.21 (1,221.21 s)[c]	1,433,128	1.43 GB (1.37 GiB)

Benchmark Suite	Workload / Cohort	Threads	Caller / Engine	Replicate Wall Times (s)[a]	Wall Mean (Min–Max) (s)[a]	Replicate Peak RSS (KB)[a]	Peak RSS Mean (MB / GiB)
Full-Genome WGS Scaling (chr1–chr22)[c]	GIAB HG008 [19] 364x Full WGS (22 Unchunked Chromosomes, -U)	32	VarDictRust	10192.00	2:49:52 (10,192.00 s)[c]	12,135,724	12.14 GB (11.57 GiB)
Full-Genome WGS Scaling (chr1–chr22)[c]	GIAB HG008 [19] 364x Full WGS (≤50 kbp GIAB Callable BED, -U)	32	VarDictRust	6355.00	1:45:55 (6,355.00 s)[c]	4,877,136	4.88 GB (4.65 GiB)

[a]Unless otherwise noted, rows report all n=3 technical replicate measurements along with the replicate mean (min–max) on GCP n2-standard-32 (Intel Xeon Platinum 8373C Ice Lake) instances, measured via /usr/bin/time -v. [b]The 7-caller somatic comparison on SRR15006376 / SRR15006386 [11] profiles single-run 8-thread executions across VarDictRust, VarDictJava [1], vardictcpp [11], Strelka2 [4], LoFreq [7], Octopus [5], and Mutect2 [8]. [c]Full-genome 32-thread WGS runs on GIAB HG002 (130.8 GB) [13, 18] and GIAB HG008 (329.0 GB, 364x combined depth) [19] report single-run wall-clock times as MM:SS.ss or H:MM:SS (with total seconds in parentheses).

Supplementary Table S4. Verified Truth-Set Operating Points, Variant-Class Stratification & 18-Arm Somatic Noise-Floor Matrix

Benchmark Suite	Cohort / Platform	Stratum / Regime / Operating Point	Caller / Parity Status	Primary Metric 1 (TP / Candidates / SNV)	Primary Metric 2 (FP / PASS / InDel)	Summary Score / F1 [12]
QUAL ROC Operating Points[a]	GIAB HG002 GRCh38 [13, 18]	All-PASS (Unfiltered, -f 0.01)	VarDictRust == VarDictJava [1]	TP=23,062	FP=22,458, FN=813	P=0.5038, R=0.9659, F1=0.6622
QUAL ROC Operating Points[a]	GIAB HG002 GRCh38 [13, 18]	Optimal QUAL (≥ 128.0)	VarDictRust == VarDictJava [1]	TP=23,004	FP=553, FN=871	P=0.9763, R=0.9635, F1=0.9699
QUAL ROC Operating Points[a]	SEQC2 HCC1395 [10]	All-PASS (Unfiltered, -f 0.01)	VarDictRust == VarDictJava [1]	TP=1,069	FP=11,334, FN=140	P=0.0852, R=0.8842, F1=0.1555
QUAL ROC Operating Points[a]	SEQC2 HCC1395 [10]	Optimal QUAL (≥ 126.0)	VarDictRust == VarDictJava [1]	TP=794	FP=293, FN=415	P=0.7279, R=0.6567, F1=0.6905
QUAL ROC Operating Points[a]	GIAB HG008 v0.3 [19]	All-PASS (Unfiltered, -f 0.01)	VarDictRust == VarDictJava [1]	TP=619	FP=1,565, FN=288	P=0.2687, R=0.6825, F1=0.3856
QUAL ROC Operating Points[a]	GIAB HG008 v0.3 [19]	Optimal QUAL (≥ 176.0)	VarDictRust == VarDictJava [1]	TP=593	FP=38, FN=314	P=0.9354, R=0.6538, F1=0.7696
QUAL ROC Operating Points[a]	GIAB HG005 GRCh38 [13]	All-PASS (Unfiltered, -f 0.01)	VarDictRust == VarDictJava [1]	TP=29	FP=2, FN=1	P=0.9355, R=0.9667, F1=0.9508
QUAL ROC Operating Points[a]	GIAB HG005 GRCh38 [13]	Optimal QUAL (≥ 226.0)	VarDictRust == VarDictJava [1]	TP=29	FP=1, FN=1	P=0.9667, R=0.9667, F1=0.9667
QUAL ROC Operating Points[a]	GIAB NA24385 GRCh37 [13]	All-PASS (Unfiltered, -f 0.01)	VarDictRust == VarDictJava [1]	TP=5	FP=81, FN=0	P=0.0581, R=1.0000, F1=0.1099
QUAL ROC Operating Points[a]	GIAB NA24385 GRCh37 [13]	Optimal QUAL (≥ 160.0)	VarDictRust == VarDictJava [1]	TP=5	FP=0, FN=0	P=1.0000, R=1.0000, F1=1.0000
Variant-Class Stratification[b]	GIAB HG002 GRCh38 [13, 18]	Variant-Class & Allele-Frequency Breakdown	VarDictRust	SNV=209,030, Ins=2,305, Del=10,975	MNP=2,065, SV=3,187	PASS=48,447 / Total=227,562
Variant-Class Stratification[b]	GIAB HG002 GRCh38 [13, 18]	Variant-Class & Allele-Frequency Breakdown	VarDictJava [1]	SNV=209,030, Ins=2,305, Del=10,975	MNP=2,065, SV=3,187	PASS=48,447 / Total=227,562
Variant-Class Stratification[b]	GIAB HG002 GRCh38 [13, 18]	Variant-Class & Allele-Frequency Breakdown	vardictcpp [11]	SNV=209,066, Ins=2,309, Del=10,949	MNP=2,049, SV=3,187	PASS=48,431 / Total=227,560
Variant-Class Stratification[b]	SEQC2 HCC1395 [10]	Variant-Class & Allele-Frequency Breakdown	VarDictRust	SNV=594,344, Ins=8,966, Del=15,716	MNP=4,383, SV=35	PASS=120,307 / Total=623,444
Variant-Class Stratification[b]	SEQC2 HCC1395 [10]	Variant-Class & Allele-Frequency Breakdown	VarDictJava [1]	SNV=594,344, Ins=8,966, Del=15,716	MNP=4,383, SV=35	PASS=120,307 / Total=623,444
Variant-Class Stratification[b]	SEQC2 HCC1395 [10]	Variant-Class & Allele-Frequency Breakdown	vardictcpp [11]	SNV=594,344, Ins=8,966, Del=15,715	MNP=4,383, SV=34	PASS=120,305 / Total=623,442
Variant-Class Stratification[b]	GIAB HG008 v0.3 [19]	Variant-Class & Allele-Frequency Breakdown	VarDictRust	SNV=15,213, Ins=1,161, Del=1,410	MNP=614, SV=28	PASS=11,358 / Total=18,426
Variant-Class Stratification[b]	GIAB HG008 v0.3 [19]	Variant-Class & Allele-Frequency Breakdown	VarDictJava [1]	SNV=15,213, Ins=1,161, Del=1,410	MNP=614, SV=28	PASS=11,358 / Total=18,426
Variant-Class Stratification[b]	GIAB HG008 v0.3 [19]	Variant-Class & Allele-Frequency Breakdown	vardictcpp [11]	SNV=15,214, Ins=1,161, Del=1,408	MNP=614, SV=28	PASS=11,356 / Total=18,425
18-Arm Somatic Noise Floor[c]	Element AVITI 111x	Min VAF 1%, ≥2 Reads (-f 0.01 -r 2)	100% Exact Match (VarDictRust == VarDictJava [1])	Total Candidates=61	PASS Somatic=13	0 differences vs. VarDictJava (eps=1e-6)[c]

Benchmark Suite	Cohort / Platform	Stratum / Regime / Operating Point	Caller / Parity Status	Primary Metric 1 (TP / Candidates / SNV)	Primary Metric 2 (FP / PASS / InDel)	Summary Score / F1 [12]
18-Arm Somatic Noise Floor[c]	Element AVITI 111x	All VAFs, ≥2 Reads (-f 0 -r 2)	100% Exact Match (VarDictRust == VarDictJava [1])	Total Candidates=61	PASS Somatic=13	0 differences vs. VarDictJava (eps=1e-6)[c]
18-Arm Somatic Noise Floor[c]	Element AVITI 111x	Singleton Noise Floor (-f 0 -r 1)	100% Exact Match (VarDictRust == VarDictJava [1])	Total Candidates=186	PASS Somatic=13	0 differences vs. VarDictJava (eps=1e-6)[c]
18-Arm Somatic Noise Floor[c]	PacBio Onso 136x	Min VAF 1%, ≥2 Reads (-f 0.01 -r 2)	100% Exact Match (VarDictRust == VarDictJava [1])	Total Candidates=76	PASS Somatic=19	0 differences vs. VarDictJava (eps=1e-6)[c]
18-Arm Somatic Noise Floor[c]	PacBio Onso 136x	All VAFs, ≥2 Reads (-f 0 -r 2)	100% Exact Match (VarDictRust == VarDictJava [1])	Total Candidates=78	PASS Somatic=19	0 differences vs. VarDictJava (eps=1e-6)[c]
18-Arm Somatic Noise Floor[c]	PacBio Onso 136x	Singleton Noise Floor (-f 0 -r 1)	100% Exact Match (VarDictRust == VarDictJava [1])	Total Candidates=185	PASS Somatic=19	0 differences vs. VarDictJava (eps=1e-6)[c]
18-Arm Somatic Noise Floor[c]	Illumina 195x	Min VAF 1%, ≥2 Reads (-f 0.01 -r 2)	100% Exact Match (VarDictRust == VarDictJava [1])	Total Candidates=115	PASS Somatic=24	0 differences vs. VarDictJava (eps=1e-6)[c]
18-Arm Somatic Noise Floor[c]	Illumina 195x	All VAFs, ≥2 Reads (-f 0 -r 2)	100% Exact Match (VarDictRust == VarDictJava [1])	Total Candidates=131	PASS Somatic=24	0 differences vs. VarDictJava (eps=1e-6)[c]
18-Arm Somatic Noise Floor[c]	Illumina 195x	Singleton Noise Floor (-f 0 -r 1)	100% Exact Match (VarDictRust == VarDictJava [1])	Total Candidates=1,199	PASS Somatic=23	0 differences vs. VarDictJava (eps=1e-6)[c]
18-Arm Somatic Noise Floor[c]	Ultima ppmSeq 157x	Min VAF 1%, ≥2 Reads (-f 0.01 -r 2)	100% Exact Match (VarDictRust == VarDictJava [1])	Total Candidates=574	PASS Somatic=133	0 differences vs. VarDictJava (eps=1e-6)[c]
18-Arm Somatic Noise Floor[c]	Ultima ppmSeq 157x	All VAFs, ≥2 Reads (-f 0 -r 2)	100% Exact Match (VarDictRust == VarDictJava [1])	Total Candidates=581	PASS Somatic=136	0 differences vs. VarDictJava (eps=1e-6)[c]
18-Arm Somatic Noise Floor[c]	Ultima ppmSeq 157x	Singleton Noise Floor (-f 0 -r 1)	100% Exact Match (VarDictRust == VarDictJava [1])	Total Candidates=2,791	PASS Somatic=159	0 differences vs. VarDictJava (eps=1e-6)[c]
18-Arm Somatic Noise Floor[c]	Roche SBX 119x	Min VAF 1%, ≥2 Reads (-f 0.01 -r 2)	100% Exact Match (VarDictRust == VarDictJava [1])	Total Candidates=770	PASS Somatic=293	0 differences vs. VarDictJava (eps=1e-6)[c]
18-Arm Somatic Noise Floor[c]	Roche SBX 119x	All VAFs, ≥2 Reads (-f 0 -r 2)	100% Exact Match (VarDictRust == VarDictJava [1])	Total Candidates=788	PASS Somatic=295	0 differences vs. VarDictJava (eps=1e-6)[c]
18-Arm Somatic Noise Floor[c]	Roche SBX 119x	Singleton Noise Floor (-f 0 -r 1)	100% Exact Match (VarDictRust == VarDictJava [1])	Total Candidates=3,354	PASS Somatic=296	0 differences vs. VarDictJava (eps=1e-6)[c]
18-Arm Somatic Noise Floor[c]	Ultima UG100 306x	Min VAF 1%, ≥2 Reads (-f 0.01 -r 2)	100% Exact Match (VarDictRust == VarDictJava [1])	Total Candidates=944	PASS Somatic=376	0 differences vs. VarDictJava (eps=1e-6)[c]
18-Arm Somatic Noise Floor[c]	Ultima UG100 306x	All VAFs, ≥2 Reads (-f 0 -r 2)	100% Exact Match (VarDictRust == VarDictJava [1])	Total Candidates=1,519	PASS Somatic=472	0 differences vs. VarDictJava (eps=1e-6)[c]
18-Arm Somatic Noise Floor[c]	Ultima UG100 306x	Singleton Noise Floor (-f 0 -r 1)	100% Exact Match (VarDictRust == VarDictJava [1])	Total Candidates=4,994	PASS Somatic=538	0 differences vs. VarDictJava (eps=1e-6)[c]
Chr20 Exome 7-Caller Benchmark	GIAB HG002 Chr20 Exome [11, 13, 20]	7-Caller Truth Benchmark	VarDictRust == VarDictJava [1]	Calls=7,422	SNV F1=0.9858	InDel F1=0.8956 (see Table 3)
Native GRCh38 Caller Matrix	GIAB HG002 [13], SEQC2 [10] & HG008 [19] GRCh38	19-Run Native GRCh38 Caller Matrix	VarDictRust == VarDictJava [1]	HG002 SNV F1=0.6619	HG002 InDel F1=0.7177	100% Exact Match (see Table 3)
Low-VAF Tumor-Purity Titration	GIAB HG008 [19] Tumor-Purity Titration (20% to 1%)	30-Run Low-VAF Somatic Curve (-f 0.002)	VarDictRust == VarDictJava [1]	20% Purity SNV F1=0.7512	1% Purity SNV F1=0.0569	100% Exact Match across all 5 purities (see Table 3)
Next-Gen Germline & Mosaic Suite	GIAB HG002 v4.2.1 [13], v5.0q T2T-Q100 [14, 16], CMRG [15] & Mosaic [19]	12-Evaluation Next-Gen Germline & Mosaic Suite	Full Contigs (Rust) & ≤50 kbp BED (VarDictRust == VarDictJava [1])	v5.0q SNV F1=0.6650 (Full) / 0.6461 (BED)	CMRG SNV F1=0.6791 (Full) / 0.6132 (BED)	Mosaic Recall=98.82% (see Table 7)
HG008 v0.4 Somatic Landscape	GIAB HG008 v0.4 Somatic Benchmark [19]	11-Caller Modern ML & Commercial Landscape	VarDictRust v1.0.6 Full WGS & Slice	Full WGS SNV F1=0.7886	Full WGS InDel F1=0.4898	27 evaluations across 11 callers (see Table 7)

[a]VarDictRust and VarDictJava v1.8.3 [1] produce identical rtg vcfeval [12] TP, FP, FN, precision, recall, and F1 scores at every QUAL threshold from 0 to 200 across all 5 truth cohorts. [b]Variant-class stratification compares VarDictRust, VarDictJava [1], and vardictcpp [11] across SNVs, insertions, deletions, complex MNPs, and structural variants (SVs), showing that vardictcpp splits complex MNPs on HG002 and omits SVs on SEQC2 and HG008 (CPP-2, CPP-9). [c]Across all 18/18 targeted Chromosome 20 somatic noise-floor runs (6 HG008 platforms x 3 low-VAF regimes: -f 0.01 -r 2, -f 0 -r 2, and singleton -f 0 -r 1), VarDictRust and VarDictJava emit identical candidate and PASS variant sets.

Supplementary Table S5. Complete 70-Cell GA4GH v3.5 [17] Genomic Stratification Matrix (7 Strata x 10 Sequencing Platforms)

Cohort	Sequencing Platform[a]	GA4GH v3.5 Stratum [17][b]	SNV Precision [12]	SNV Recall [12]	SNV F1 [12]	InDel Precision / Recall / F1 [12][c]
GIAB HG002 Germline [13, 18]	Illumina	Homopolymer (4–6 bp)	0.4869	0.9645	0.6471	0.4874 / 0.9134 / 0.6356
GIAB HG002 Germline [13, 18]	Illumina	Homopolymer (7–11 bp)	0.6056	0.9278	0.7328	0.6079 / 0.9474 / 0.7406
GIAB HG002 Germline [13, 18]	Illumina	Homopolymer (≥12 bp)	0.6828	0.8704	0.7653	0.8869 / 0.7491 / 0.8122
GIAB HG002 Germline [13, 18]	Illumina	Tandem Repeats	0.5462	0.9126	0.6833	0.8782 / 0.8357 / 0.8564
GIAB HG002 Germline [13, 18]	Illumina	Segmental Duplications	0.5343	0.8980	0.6700	0.5650 / 0.8514 / 0.6792
GIAB HG002 Germline [13, 18]	Illumina	Low Mappability	0.5202	0.8309	0.6398	0.4419 / 0.8199 / 0.5742
GIAB HG002 Germline [13, 18]	Illumina	RefSeq CDS	0.3644	0.9831	0.5317	0.1209 / 1.0000 / 0.2157
GIAB HG002 Germline [13, 18]	Element AVITI	Homopolymer (4–6 bp)	0.9491	0.9685	0.9587	0.9371 / 0.9308 / 0.9339
GIAB HG002 Germline [13, 18]	Element AVITI	Homopolymer (7–11 bp)	0.9657	0.9325	0.9488	0.9626 / 0.9533 / 0.9579
GIAB HG002 Germline [13, 18]	Element AVITI	Homopolymer (≥12 bp)	0.9793	0.8765	0.9251	0.9882 / 0.8968 / 0.9403
GIAB HG002 Germline [13, 18]	Element AVITI	Tandem Repeats	0.8662	0.9383	0.9008	0.9430 / 0.8396 / 0.8883
GIAB HG002 Germline [13, 18]	Element AVITI	Segmental Duplications	0.7358	0.8980	0.8088	0.7485 / 0.8446 / 0.7937
GIAB HG002 Germline [13, 18]	Element AVITI	Low Mappability	0.6786	0.8419	0.7515	0.6406 / 0.7640 / 0.6969
GIAB HG002 Germline [13, 18]	Element AVITI	RefSeq CDS	0.9311	0.9887	0.9591	0.7857 / 1.0000 / 0.8800
GIAB HG002 Germline [13, 18]	MGI DNBSEQ	Homopolymer (4–6 bp)	0.5416	0.9632	0.6934	0.4645 / 0.9279 / 0.6191
GIAB HG002 Germline [13, 18]	MGI DNBSEQ	Homopolymer (7–11 bp)	0.5844	0.9284	0.7173	0.2319 / 0.9519 / 0.3729
GIAB HG002 Germline [13, 18]	MGI DNBSEQ	Homopolymer (≥12 bp)	0.4531	0.8796	0.5981	0.7337 / 0.8764 / 0.7987
GIAB HG002 Germline [13, 18]	MGI DNBSEQ	Tandem Repeats	0.4990	0.9326	0.6502	0.6606 / 0.8336 / 0.7371
GIAB HG002 Germline [13, 18]	MGI DNBSEQ	Segmental Duplications	0.5267	0.8960	0.6634	0.4792 / 0.8581 / 0.6150
GIAB HG002 Germline [13, 18]	MGI DNBSEQ	Low Mappability	0.3691	0.8269	0.5104	0.3342 / 0.7888 / 0.4695
GIAB HG002 Germline [13, 18]	MGI DNBSEQ	RefSeq CDS	0.4110	0.9831	0.5796	0.2703 / 0.9091 / 0.4167
GIAB HG002 Germline [13, 18]	Ultima UG100	Homopolymer (4–6 bp)	0.8208	0.9409	0.8768	0.0195 / 0.7546 / 0.0381
GIAB HG002 Germline [13, 18]	Ultima UG100	Homopolymer (7–11 bp)	0.5572	0.7097	0.6243	0.0514 / 0.4792 / 0.0928
GIAB HG002 Germline [13, 18]	Ultima UG100	Homopolymer (≥12 bp)	0.2666	0.6944	0.3853	0.2957 / 0.4573 / 0.3591
GIAB HG002 Germline [13, 18]	Ultima UG100	Tandem Repeats	0.7497	0.9100	0.8221	0.2424 / 0.7895 / 0.3709
GIAB HG002 Germline [13, 18]	Ultima UG100	Segmental Duplications	0.8726	0.8755	0.8740	0.0277 / 0.7297 / 0.0535
GIAB HG002 Germline [13, 18]	Ultima UG100	Low Mappability	0.8497	0.7799	0.8133	0.0359 / 0.6708 / 0.0682
GIAB HG002 Germline [13, 18]	Ultima UG100	RefSeq CDS	0.7489	0.9568	0.8402	0.0016 / 0.7273 / 0.0033
GIAB HG008 Somatic [19]	Illumina	Homopolymer (4–6 bp)	0.9545	0.8750	0.9130	0.7778 / 0.4118 / 0.5385
GIAB HG008 Somatic [19]	Illumina	Homopolymer (7–11 bp)	1.0000	1.0000	1.0000	1.0000 / 0.7647 / 0.8667
GIAB HG008 Somatic [19]	Illumina	Homopolymer (≥12 bp)	0.8000	0.5000	0.6154	0.8235 / 0.3373 / 0.4786
GIAB HG008 Somatic [19]	Illumina	Tandem Repeats	0.1296	0.6364	0.2154	0.0563 / 0.2105 / 0.0889
GIAB HG008 Somatic [19]	Illumina	Segmental Duplications	0.9000	0.7500	0.8182	0.8333 / 0.5556 / 0.6667
GIAB HG008 Somatic [19]	Illumina	Low Mappability	0.3158	1.0000	0.4800	0.3000 / 1.0000 / 0.4615
GIAB HG008 Somatic [19]	Illumina	RefSeq CDS	1.0000	1.0000	1.0000	No Truth InDels in Stratum[c]
GIAB HG008 Somatic [19]	Element AVITI	Homopolymer (4–6 bp)	0.8958	0.8958	0.8958	0.9375 / 0.8824 / 0.9091
GIAB HG008 Somatic [19]	Element AVITI	Homopolymer (7–11 bp)	1.0000	1.0000	1.0000	1.0000 / 0.8824 / 0.9375
GIAB HG008 Somatic [19]	Element AVITI	Homopolymer (≥12 bp)	0.8000	0.5000	0.6154	0.9500 / 0.6867 / 0.7972
GIAB HG008 Somatic [19]	Element AVITI	Tandem Repeats	0.0909	0.6364	0.1591	0.1154 / 0.3158 / 0.1690
GIAB HG008 Somatic [19]	Element AVITI	Segmental Duplications	0.7143	0.8333	0.7692	0.8333 / 0.5556 / 0.6667
GIAB HG008 Somatic [19]	Element AVITI	Low Mappability	0.2683	0.9167	0.4151	0.2727 / 1.0000 / 0.4286
GIAB HG008 Somatic [19]	Element AVITI	RefSeq CDS	1.0000	1.0000	1.0000	No Truth InDels in Stratum[c]
GIAB HG008 Somatic [19]	PacBio Onso	Homopolymer (4–6 bp)	0.9286	0.8125	0.8667	0.6667 / 0.7059 / 0.6857
GIAB HG008 Somatic [19]	PacBio Onso	Homopolymer (7–11 bp)	0.7500	0.7500	0.7500	0.8125 / 0.7647 / 0.7879
GIAB HG008 Somatic [19]	PacBio Onso	Homopolymer (≥12 bp)	0.8000	0.5000	0.6154	0.9231 / 0.5783 / 0.7111
GIAB HG008 Somatic [19]	PacBio Onso	Tandem Repeats	0.0976	0.7273	0.1720	0.0625 / 0.1579 / 0.0896
GIAB HG008 Somatic [19]	PacBio Onso	Segmental Duplications	0.6000	0.7500	0.6667	0.6667 / 0.4444 / 0.5333

Cohort	Sequencing Platform[a]	GA4GH v3.5 Stratum [17][b]	SNV Precision [12]	SNV Recall [12]	SNV F1 [12]	InDel Precision / Recall / F1 [12][c]
GIAB HG008 Somatic [19]	PacBio Onso	Low Mappability	0.2353	1.0000	0.3810	0.2143 / 1.0000 / 0.3529
GIAB HG008 Somatic [19]	PacBio Onso	RefSeq CDS	1.0000	1.0000	1.0000	No Truth InDels in Stratum[c]
GIAB HG008 Somatic [19]	Roche SBX	Homopolymer (4–6 bp)	0.8571	0.8750	0.8660	0.4000 / 0.2353 / 0.2963
GIAB HG008 Somatic [19]	Roche SBX	Homopolymer (7–11 bp)	0.8000	1.0000	0.8889	0.7000 / 0.4118 / 0.5185
GIAB HG008 Somatic [19]	Roche SBX	Homopolymer (≥12 bp)	1.0000	0.5000	0.6667	0.3913 / 0.1084 / 0.1698
GIAB HG008 Somatic [19]	Roche SBX	Tandem Repeats	0.7143	0.4545	0.5556	0.5000 / 0.1053 / 0.1739
GIAB HG008 Somatic [19]	Roche SBX	Segmental Duplications	0.8889	0.6667	0.7619	0.7500 / 0.3333 / 0.4615
GIAB HG008 Somatic [19]	Roche SBX	Low Mappability	0.7857	0.9167	0.8462	0.5000 / 1.0000 / 0.6667
GIAB HG008 Somatic [19]	Roche SBX	RefSeq CDS	1.0000	1.0000	1.0000	No Truth InDels in Stratum[c]
GIAB HG008 Somatic [19]	Ultima ppmSeq	Homopolymer (4–6 bp)	0.8667	0.8125	0.8387	0.2857 / 0.1176 / 0.1667
GIAB HG008 Somatic [19]	Ultima ppmSeq	Homopolymer (7–11 bp)	1.0000	0.2500	0.4000	0.0000 / 0.0000 / 0.0000
GIAB HG008 Somatic [19]	Ultima ppmSeq	Homopolymer (≥12 bp)	1.0000	0.1250	0.2222	1.0000 / 0.0241 / 0.0471
GIAB HG008 Somatic [19]	Ultima ppmSeq	Tandem Repeats	0.2500	0.6364	0.3590	0.0435 / 0.0526 / 0.0476
GIAB HG008 Somatic [19]	Ultima ppmSeq	Segmental Duplications	0.7273	0.6667	0.6957	0.5000 / 0.2222 / 0.3077
GIAB HG008 Somatic [19]	Ultima ppmSeq	Low Mappability	0.6154	0.6667	0.6400	1.0000 / 0.3333 / 0.5000
GIAB HG008 Somatic [19]	Ultima ppmSeq	RefSeq CDS	1.0000	1.0000	1.0000	No Truth InDels in Stratum[c]
GIAB HG008 Somatic [19]	Ultima UG100	Homopolymer (4–6 bp)	0.9512	0.8125	0.8764	0.5000 / 0.1176 / 0.1905
GIAB HG008 Somatic [19]	Ultima UG100	Homopolymer (7–11 bp)	1.0000	0.2500	0.4000	0.0000 / 0.0000 / 0.0000
GIAB HG008 Somatic [19]	Ultima UG100	Homopolymer (≥12 bp)	1.0000	0.1250	0.2222	0.2308 / 0.0361 / 0.0625
GIAB HG008 Somatic [19]	Ultima UG100	Tandem Repeats	0.6250	0.4545	0.5263	0.1429 / 0.0526 / 0.0769
GIAB HG008 Somatic [19]	Ultima UG100	Segmental Duplications	1.0000	0.8333	0.9091	1.0000 / 0.2222 / 0.3636
GIAB HG008 Somatic [19]	Ultima UG100	Low Mappability	0.9167	0.9167	0.9167	0.5000 / 0.3333 / 0.4000
GIAB HG008 Somatic [19]	Ultima UG100	RefSeq CDS	1.0000	1.0000	1.0000	No Truth InDels in Stratum[c]

[a]Evaluates Chromosome 20 callsets at -f 0.01 across 4 GIAB HG002 [13, 18] germline platforms and 6 GIAB HG008 [19] somatic platforms; across all 70/70 stratification cells (Clopper-Pearson 95% CI 94.87%–100.00%), VarDictRust v1.0.6 and VarDictJava v1.8.3 [1] exhibit 0 site differences. [b]Genomic strata follow the NIST GIAB / GA4GH v3.5 [17] stratification BED definitions on GRCh38 (homopolymers of 4–6 bp, 7–11 bp, and >=12 bp, all tandem repeats, segmental duplications, low-mappability regions, and RefSeq coding sequences). [c]Within the Chromosome 20 RefSeq CDS stratum of GIAB HG008 v0.4 [19], the somatic truth set contains SNVs only and zero truth InDels.

Supplementary Table S6. 13-Mode Functional Parity Matrix and CPP-1 to CPP-10 vardictcpp [11] Divergence Taxonomy

Evaluation Category	CLI Mode / Divergence Class	VarDictJava v1.8.3 [1] Rows / Behavior	VarDictRust v1.0.6 Rows / Behavior	vardictcpp [11] Rows / Behavior	Verification Verdict & Root Cause
13-Mode Functional Parity Suite[a]	Simple Germline (-f 0.01)	228	228 (FP=0, FN=0)	228 (FP=0, FN=0)	100% Exact Match (VarDictRust == VarDictJava)
13-Mode Functional Parity Suite[a]	Simple Zero-Threshold MRD (-f 0)	248	248 (FP=0, FN=0)	248 (FP=0, FN=0)	100% Exact Match (VarDictRust == VarDictJava)
13-Mode Functional Parity Suite[a]	Simple Chunked Mode (--chunk 50,000)	228	228 (FP=0, FN=0)	228 (FP=0, FN=0)	100% Exact Match (VarDictRust == VarDictJava)
13-Mode Functional Parity Suite[a]	Simple Germline with Native Fisher (--fisher)[a]	650	650 (FP=0, FN=0)	650 (FP=1, FN=1)	100% Exact Match (VarDictRust == VarDictJava)
13-Mode Functional Parity Suite[a]	Targeted Clinical Panel (Chr17)	83	83 (FP=0, FN=0)	83 (FP=0, FN=0)	100% Exact Match (VarDictRust == VarDictJava)
13-Mode Functional Parity Suite[a]	Structural Variant Deletion Regions	3	3 (FP=0, FN=0)	3 (FP=0, FN=0)	100% Exact Match (VarDictRust == VarDictJava)
13-Mode Functional Parity Suite[a]	Low-Coverage Boundary Calling	146	146 (FP=0, FN=0)	146 (FP=0, FN=0)	100% Exact Match (VarDictRust == VarDictJava)
13-Mode Functional Parity Suite[a]	Amplicon Calling (-a, Chr17)	145	145 (FP=0, FN=0)	145 (FP=0, FN=0)	100% Exact Match (VarDictRust == VarDictJava)
13-Mode Functional Parity Suite[a]	Amplicon Calling with Native Fisher (-a --fisher)	145	145 (FP=0, FN=0)	145 (FP=0, FN=0)	100% Exact Match (VarDictRust == VarDictJava)
13-Mode Functional Parity Suite[a]	Paired Tumor-Normal Somatic (-f 0.01)	664	664 (FP=0, FN=0)	664 (FP=0, FN=0)	100% Exact Match (VarDictRust == VarDictJava)
13-Mode Functional Parity Suite[a]	Paired Somatic Zero-Threshold MRD (-f 0)	1,396	1,396 (FP=0, FN=0)	1,396 (FP=0, FN=0)	100% Exact Match (VarDictRust == VarDictJava)
13-Mode Functional Parity Suite[a]	Paired Somatic with Native Fisher (--fisher)	527	527 (FP=0, FN=0)	527 (FP=0, FN=0)	100% Exact Match (VarDictRust == VarDictJava)

Evaluation Category	CLI Mode / Divergence Class	VarDictJava v1.8.3 [1] Rows / Behavior	VarDictRust v1.0.6 Rows / Behavior	vardictcpp [11] Rows / Behavior	Verification Verdict & Root Cause
13-Mode Functional Parity Suite[a]	Lossless CRAM-vs-BAM Twin (-G) [26][b]	Fails (NoClassDefFoundError on CRAM)[b]	228 (FP=0, FN=0)	228 (FP=0, FN=0)	100% Exact Match vs. BAM Twin[b]
vardictcpp Divergence Taxonomy [11][c]	CPP-1: VCF Header & INFO/END Schema	Standard VCF 4.2 + INFO/END tag	Exact v1.8.3 header & INFO/END parity	Malformed or omitted INFO/END & header lines	Schema mismatch in downstream VCF parsers
vardictcpp Divergence Taxonomy [11][c]	CPP-2: CRISPR (-J/-j) & Structural Variant Modes	Supported	Supported (100% Exact Match)	Unimplemented / drops SV calls	Verified across 17-modality extended coverage suite
vardictcpp Divergence Taxonomy [11][c]	CPP-3: Amplicon (-a) Output Format	38/40-col .var + VCF via var2vcf_valid.pl	Exact 38/40-col .var + VCF parity	Emits raw intermediate TSV instead of VCF	Incompatible with standard VCF comparators & bcftools [26]
vardictcpp Divergence Taxonomy [11][c]	CPP-4: Concatenated CLI Flag Parsing	Supported	Supported (60-flag CLI parity)	Exits with code 120 on bundled CLI flags	Breaks existing VarDict pipeline CLI wrappers
vardictcpp Divergence Taxonomy [11][c]	CPP-5: Chunk Boundary Deduplication (--chunk 50,000)	Single-region calling (or OOM on contigs)	100 kbp halo tiling + cross-tile dedup	Emits +309,569 duplicate boundary calls on unchunked HG002 WGS	Uncoordinated 50 kbp window splitting without dedup
vardictcpp Divergence Taxonomy [11][c]	CPP-6: Missing ##contig VCF Header Lines	Emits ##contig lines from reference .fai	Emits exact ##contig lines from reference .fai	Omits ##contig header lines	Requires manual header injection prior to rtg vcfeval [12]
vardictcpp Divergence Taxonomy [11][c]	CPP-7: GRCh37 / b37 Numeric Chromosome Naming	Supported	Supported (100% Match on GIAB NA24385 GRCh37 [13])	Hardcoded chr-prefix assumptions	Verified on GIAB NA24385 GRCh37 v4.2.1 benchmark [13]
vardictcpp Divergence Taxonomy [11][c]	CPP-8: BAM/CRAM [26] Reference & Decode	BAM (CRAM requires external htsjdk classpath)	Native BAM + CRAM (-G) zero-difference parity	Fails or diverges on CRAM twin alignments	Verified on GIAB HG008 [19] & Ultima CRAM twins
vardictcpp Divergence Taxonomy [11][c]	CPP-9: Complex MNP Splitting & Somatic SV Drops	Reference pileup & local realigner	Exact parity with VarDictJava [1]	Differs on 394 callable HG002 WGS calls & drops SVs	Local realignment & tie-break divergence vs. oracle
vardictcpp Divergence Taxonomy [11][c]	CPP-10: Transcriptome RNA-Seq Splice Mode (-i)	Supported (12,668 rows on HG002 mRNA [13])	Supported (12,668 rows, 100% Exact Match)	Unimplemented (unsupported flag)	Verified on 51.8 GiB GIAB HG002 GM24385 STAR BAM [13]

[a]Evaluates all 13 functional CLI modes across VarDictJava v1.8.3 [1], VarDictRust v1.0.6, and vardictcpp [11]; VarDictRust achieves 0 false positives (FP=0) and 0 false negatives (FN=0) across all 13/13 modes, whereas vardictcpp exhibits FP=1, FN=1 on simple_fisher. [b]Stock Bioconda VarDictJava v1.8.3 [1] fails with java.lang.NoClassDefFoundError on CRAM inputs; VarDictRust decodes CRAM natively via pure-Rust noodles and matches the lossless BAM twin oracle with 0 differences. [c]Catalogs the 10 functional, schema, and algorithmic divergence classes (CPP-1 through CPP-10) empirically observed in vardictcpp (commit ae93fe39dc) [11] during the 60-experiment benchmark study.

6. References

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